PRESIDENCY UNIVERSITY

Private University Estd. in Karnataka State by Act No. 41 of 2013

Itgalpura, Rajankunte, Yelahanka, Bengaluru – 560064

Millets Value Chain Platform
A Digital Marketplace for "Shree Anna"

A PROJECT REPORT

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Certified that this report "MILLETS VALUE CHAIN PLATFORM A DIGITAL MARKETPLACE FOR "SHREE ANNA"" is a Bonafide work of "TEJASWINI B (20221CSE0342), INDUJA R (20221CSE0317), M NAVYASHREE (20221CSE0347)", who have successfully carried out the project work and submitted the report for partial fulfilment of the requirements for the award of the degree of BACHELOR OF TECHNOLOGY in COMPUTER SCIENCE ENGINEERING during 2025-26.

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DECLARATION

We the students of final year B. Tech in COMPUTER SCIENCE ENGINEERING at Presidency University, Bengaluru, named TEJASWINI B, INDUJA R AND M NAVYASHREE, hereby declare that the project work titled **"MILLETS VALUE CHAIN PLATFORM A DIGITAL MARKETPLACE FOR "SHREE ANNA""** has been independently carried out by us and submitted in partial fulfillment for the award of the degree of B.Tech in COMPUTER SCIENCE ENGINEERING during the academic year of 2025-26. Further, the matter embodied in the project has not been submitted previously by anybody for the award of any Degree or Diploma to any other institution.

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Abstract

The nutritional value of the millets has provided them with a renewed significance in the world, along with their climate and environmental sustainability in agriculture. However, the existing millet supply chain in developing countries, including India, is significantly fragmented. This is characterized by limited access to markets; lack of price information; and challenges in coordinating the supply chain as well as how much is known about the products from point of origin through point of end use.

In addition to the structural issues of the current supply chain, there are also issues related to lack of recognition and support from policymakers on millet crops, low-level certification systems, intermediaries, and many other factors that limit the ability of tribal and smallholder farmers to earn income or to participate fully in markets. This paper proposes a concept for a Digital Millet Supply Chain Platform that removes all financial barriers between family farmers, startup businesses, consumers, and other entities by creating one comprehensive online platform for all transactions. Examples of types of transactions that would take place on this Digital Millet Supply Chain Platform include direct-to-consumer and direct-to-startup sales; verifiable online seller status; raw material purchases from family farmers; sales of processed goods from family farmers; traceability; price communication; a waste-to-value system, i.e., ACM; and online order tracking as well as online payment tracking.

The Digital Millet Supply Chain Platform will be similar to models of other digitalized agri-food systems and traceability around the world and will be guided by the principles of modular architecture; role-based dashboards; digital monitoring; and value chain sustainability. By applying these same principles to optimize the value chain through using millet-based markets and consumer-focused products, we can create a technology-driven supply chain that will provide a common platform to enhance farmer income, develop startups, assure product quality, improve market transparency/use of supply traceability, and be sustainable in relation to the millet value chain.

The solution that has been identified will ultimately create an inclusive, efficient, and digitally-enabled agriculture market aligned with the future requirements of the millet economy.

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Abbreviations

Abbreviation	Full Form
ACM	Accessories, Craft & More
API	Application Programming Interface
CSS	Cascading Style Sheets
eNAM	National Agriculture Market
ENUM	Enumeration
HTML	Hyper Text Markup Language
JSON	JavaScript Object Notation
JWT	JSON Web Tokens
MySQL	My Structured Query Language
OTP	One-time password
REST API	Representational State Transfer Application Programming Interface
SDG	Sustainable Development Goals
TTS	Text-to-Speech
UI/UX	User Interface / User Experience

Chapter 1

Introduction

India has been ranked among the largest agricultural economies across the globe with agriculture being critical in the maintenance of livelihoods, food security and rural development. Millets are one of the many crops that has been in the limelight around the world because of its high nutritional value, capacity to withstand climate change, and sustainability in agricultural practice. Some of the common varieties like ragi, jowar, bajra and foxtail millets contain the required amount of fiber, protein and other essential minerals needed in the healthy diet. Global and national efforts to promote sustainable food systems also help in the growing popularity of millets (FAO, 2023; Government of India, 2023).

The millet ecosystem in India has various challenges despite their advantages, such as the fragmented supply chains, absence of direct market linkages, poor price transparency, poor consumer awareness, and absence of technological integration in value chain process. The challenges demonstrate the necessity to digitalize the millet value chain with the help of digital technologies and intelligent farming solutions.

The Millets Value Chain Platform is created as a digital platform that is both integrated and interactive and is created to connect all the stakeholders in the millet ecosystem, such as farmers, suppliers, distributors, retailers, and consumers. The site is expected to enhance efficiency in the supply chain and transparency, traceability and availability of millet information and services. These digital interventions have been found to be crucial in enhancing agricultural value chains and coordination of the stakeholders (Tripathi, Singh and Dutta, 2022).

This chapter gives the background knowledge of the project, its background, major statistics, current technologies, the approach suggested, the aims, how it will be used in the Sustainable Development Goals (SDGs), and the structure of the entire report.

1.1 Background

India is traditionally one of the biggest millet producers. Conventional agricultural societies have been dependent on these crops because they are drought resistant, do not demand much water and they also adapt well to the changes in climatic conditions. Millets have been part of the rural diet and farming since time immemorial. Nevertheless, during the past decades, the production and the use of the millets decreased greatly with the prevalence of rice and wheat in the social distribution and in the cities in the diets. This has led to low incomes of the farmer, lack of use of agricultural land that can be used to produce millet and lack of awareness by the consumer on their advantages. Innovative issues like global warming, soil erosion, and water shortage have revived the efforts to use millets as an alternative that is sustainable.

International organizations and governments have encouraged the growing of millet including the declaration of the International Year of Millets 2023 (Government of India, 2023). Irrespective of such efforts, the millet value chain is not yet complete. Farmers do not usually have direct access to markets and rely on the intermediaries, which decreases their profit margins. The processes involved in the supply chain, including production, storage, transportation and distribution are poorly integrated and managed. The consumers are also challenged to get quality authentic millet products. The literature on agricultural value chains points out that absence of integration and coordination results in inefficiencies and diminished economic gains to the stakeholders (Trienekens, 2011).

Current agricultural platforms are mainly offering isolated services like crop information or market prices, but not a complete millet-centered solution. This brings about disintegration between production and consumption, which results in inefficiencies along the value chain. This gap can be bridged now with the development of web technologies, data management systems, and digital platforms.

The Millets Value Chain Platform aims to revolutionize and integrate the millet supply chain, enhance linkages between the farmers and the markets, encourage sustainable agriculture, and enhance consumer awareness and accessibility.

1.2 Project Statistics

The Millets Value Chain Platform is planned to be a scalable digital ecosystem that serves the needs of various stakeholders and processes in the millet supply chain. The efficiency, usability, and scalability are achieved with a structured and data-driven approach.

1.2.1 Content Statistics

The site classifies information into different categories to reflect the whole millet value chain as a whole. It contains details of large millet growing areas in India and addresses different types of millet like ragi, jowar, bajra and foxtail millet. The system keeps a high degree of records of farmers, their products, suppliers and distribution channels. It also records the pricing of products used, demand in the market and nutritional value to consumers.

Moreover, the site also offers information about millet practices of cultivation and health, making it a complete source of information to all stakeholders.

1.2.2 Technical Statistics

The technology stack enables the platform to stay optimized and be scalable:

Table 1.1 Technical Statistics

Component	**Value / Statistics**
Backend Technology	Node.js with modular API architecture
Database Structure	MySQL with 12+ relational tables
Frontend Components	100+ reusable UI components using React.js with CSS (inline styling)
Data Format	JSON for efficient data handling
Pages/Views	30+ interactive pages (Dashboard, Farmer Portal, Marketplace, etc.)
Load Time	Less than 3 seconds (optimized performance)

1.2.3 User Interaction Statistics

The portal is to serve a wide variety of users, such as farmers, suppliers, retailers, consumers, and policymakers. At the first stage, the number of users served is expected to be 5,000 to 10,000, and the backend will be able to process the requests on a daily basis. It will have the capacity to serve over 50 transactions daily at an initial point. In addition, the architecture is also scalable, which means that the platform could be expanded to the national-level agricultural system in the future. The scalability corresponds to the increasing prominence of digital marketplaces in agriculture, which allow access to a broader audience and better economic performance (Sharma and Singh, 2021).

1.3 Prior Existing Technologies

An in-depth study of the currently available digital agricultural platforms shows that they have a number of limitations even though they have brought a difference in the sector. There are numerous sites with helpful functions like market price and farmer databases, but they are not always integrated, interactive and designed in an easy way. It has been studied that digital agricultural systems often do not offer a coherent view of the value chain and do not focus on particular crops, such as millets (Tripathi, Singh and Dutta, 2022).

1.3.1 Government Agricultural Initiatives

Indian digital projects being spearheaded by the government have contributed greatly towards changing agriculture in India.

The most renowned of these platforms is the National Agriculture Market (eNAM), which links agricultural mandis nationwide and offers real-time price quotations in the market. Although eNAM has been effective in enhancing price transparency and trading at a large scale, it is not specifically oriented towards millets, but instead, it is designed to work with general agricultural commodities. Also, the system is not very interactive to the end-users as well as lacks full supply chain tracking.

Agri Stack, the other significant initiative, tries to establish a single digital ecosystem by combining databases with farmers and agricultural services. Despite the great potential of Agri Stack in their future development, a lot of its components are at an early level.

The existing implementation does not have a high level of direct user interaction and cannot be able to see the processes in the supply chain in real time.

Similarly, various platforms under the Digital Agriculture Mission provide valuable agricultural data such as crop information, weather forecasts, and advisory services. These systems are stable and readily available but they are mainly information based and do not provide an interactive environment, visualization, and integration between the various levels of the agricultural value chain. Furthermore, they are not specifically oriented on millets, which also constrains their applicability to this project.

1.3.2 Private and Semi-Governmental Platforms

Besides government programs, there have been a number of Agri-tech platforms, both private and semi-governmental, that have risen to enhance agricultural efficiency and market access. Technology-based solutions like DeHaat and Ninjacart have brought in a supply chain logistics in order to provide direct connection between farmers and buyers. These systems have highly enhanced efficiency in operations and have minimized delays in agricultural distribution systems.

Nevertheless, they mostly concentrate on crops that are of high demand and commercial dominance as opposed to millets. Consequently, not all millet farmers have equal benefits with these systems. Moreover, the platforms tend to focus on the business functions and they may not offer full transparency to all the stakeholders. Accessibility in some of the rural areas might also be hindered by technological and infrastructural factors.

Digital agricultural ecosystems are also enabled by state-level portals to agricultural, which give region-specific data and services supported by the government. Whereas these platforms provide useful localized data, they are not standardized across various states, and thus, information structures are fragmented. Moreover, they are usually administrative and lack interactive and user-friendly interfaces to farmers and consumers.

1.3.3 Digital Data repositories and Agricultural information systems

Digital libraries and agricultural data repositories are significant in storing and sharing agricultural information.

These websites offer data concerning crop production, price dynamics, soil properties, and agricultural activities. As much as they are good sources of information, the information is usually in disorganized form and hard to comprehend especially to farmers who are less technical.

The other weakness of these repositories is that they lack interactive dashboards and visualization tools that limit the capability of users to extract valuable insights out of the data. Most of the time, the user has to analyse raw data manually thereby lowering usability and effectiveness.

Global agricultural platforms, in their turn, provide more sophisticated analytics and high-quality data. Nevertheless, they are not Indian agricultural and do not have a regional orientation. Neither do they prioritize data related to millet, which makes them less relevant to this project. Therefore, their applicability by Indian farmers is not high, notwithstanding the fact that they are technologically powerful.

1.3.4 Existing System limitations

The discussion of the current platforms reveals that there are some significant constraints that impact their performance. The issue of lack of integration can be seen as one of the main problems, because most platforms concentrate on single functionalities, i.e. pricing or advisory services, but not on an end-to-end solution. This creates divided information whereby users have to use several platforms to get complete information.

The other important weakness is the uninteractivity. Most systems are only a repository of information that is not dynamic and provides fewer opportunities to engage with the system. This minimizes their use and does not promote active involvement of farmers and other stakeholders.

Access is also a key issue. Most websites are not designed to be mobile friendly or to function with low bandwidth meaning that they cannot be easily accessed in rural settings. Furthermore, the lack of multilingual support and inclusive design also limits their utilization by different groups of users.

Underrepresentation of millets is a critical gap that is observed in these systems. Although they are essential in sustainable farming, there are limited platforms that specifically support the cultivation of millet, tracking of their supply chains, or integrating them into the market. This inability to focus greatly restricts the millet ecosystem development.

These shortcomings are sufficient reasons to have a consolidated, interactive, and millet-oriented digital platform that incorporates the entire value chain into one platform.

1.4 Proposed Approach

Current agricultural platforms are prone to fragmentation, lack transparency and user interaction, which limit their effectiveness in assisting stakeholders throughout the millet value chain. The proposed Millets Value Chain Platform combats these challenges by being user-centric, technology-driven, and integrated digitally. Its main purpose is not just to digitalize the data on millet but also to turn the whole value chain into an interactive, transparent, and efficient ecosystem (Jayalath et al., 2021; Sharma and Singh, 2021).

1.4.1 Value Chain System which is centralized and organized under Millet

Information on agriculture on millets is normally spread among various stakeholders making it inefficient and inaccessible. The suggested platform provides a centralized framework that gathers the information regarding the farmers, products, supply chain phases, and the market dynamics in a systematic classification. This single system increases the accessibility of data, effective management, and enables users to navigate the information by stakeholders, thus increasing the overall coordination in the value chain (Trienekens, 2011; Banu et al., 2019).

1.4.2. Interactive, Engaging User Experience

Conventional farming processes tend to use fixed data display, which restricts interaction with the user. Conversely, the suggested platform will include interactive dashboards and real-time tracking of the supply chain along with a specific marketplace of millet products. These functionalities allow users to be actively involved in the platform, discover insights, and make informed decisions. This interactivity is also in line with the current trends in digital agriculture, where focus is placed on usability and user engagement (Tripathi et al., 2022).

1.4.3 Technology-Inspired Accessibility and Inclusiveness

The platform has accessibility and inclusivity as its design principles. It uses a highly responsive and lightweight interface built with Tailwind CSS that makes the system usable on all devices and under different networks. It is also low data consumption, and thus it can be used by rural users who lack access to the internet. Moreover, future developments like the addition of multilingual support and voice-based interaction are set to enhance inclusivity further and reduce the digital divide (OECD, 2020).

1.4.4 Scalable and Moderning Web Technologies

The platform is designed with a scalable and strong technology stack to guarantee performance and reliability. The backend is developed in Node.js and express and MySQL have been used to manage data effectively. The front is developed using HTML, Tailwind CSS, JavaScript and React.js, which allows dynamic and responsive user interfaces. The use of JSON-based data management allows flexibility and smooth communication between the system components and promotes scalability and further growth (Wolfert et al., 2017).

1.4.5 Future-Friendly and Continuous Design

The system is designed with long-term adaptability in mind. A scalable database architecture supports increasing data volumes, while modular system design allows seamless integration of new features. The maintainable code structure ensures continuous improvement and facilitates the incorporation of emerging technologies such as AI and blockchain for enhanced traceability and efficiency (Kamilaris et al., 2019).

1.4.6 User-Centric and Learning-Oriented Approach

Unlike conventional platforms that focus primarily on data display, this system emphasizes user experience and learning. It incorporates visual dashboards, simplified data representation, and informative content to enhance user understanding. This approach improves usability and encourages active participation, particularly among farmers and non-technical users (Vattekkad et al., 2020).

1.4.7 Bridging the Gap in Agricultural Value Chain

A key objective of the platform is to integrate all stakeholders—farmers, suppliers, distributors, and consumers into a unified digital ecosystem. By enabling seamless interaction and information exchange, the system enhances transparency, coordination, and efficiency across the millet value chain, addressing critical gaps identified in existing agricultural systems (Ahmed et al., 2021).

1.5 Objectives

The objectives of the project are designed to address the limitations of existing agricultural platforms while promoting sustainability, transparency, and technological advancement in the millet ecosystem.

1.5.1 Primary Objectives

The primary objective is to consolidate the fragmented millet value chain into a unified digital platform. Currently, information related to production, processing, distribution, and consumption is scattered, resulting in inefficiencies and limited coordination among stakeholders. The proposed system integrates these components into a centralized platform that provides structured and accessible information on farmer data, production processes, market availability, and commercial transactions (Ritu and Mercykutty, 2025).

Another key objective is to enhance user engagement through intuitive design and interactive features. Unlike traditional platforms that rely on static content, the proposed system incorporates dashboards, real-time updates, and stakeholder-specific modules to improve usability and accessibility.

The platform also aims to economically empower farmers by reducing dependency on intermediaries and enabling direct market access. By providing transparent pricing information and facilitating direct transactions, the system helps farmers secure better returns for their produce (Adekunle et al., 2018).

Additionally, the platform promotes awareness and consumption of millets by highlighting their nutritional and environmental benefits. This contributes to improved public health and supports sustainable agricultural practices (Dayakar Rao et al., 2017; FAO, 2023).

1.5.2 Technical Objectives

The technical objective is to develop a scalable, high-performance web platform that operates efficiently across multiple devices. The system leverages modern web technologies such as HTML, Tailwind CSS, JavaScript, and React for frontend development, while Node.js and MySQL support backend operations.

The platform emphasizes efficient API communication, optimized database queries, and modular architecture. This design enables easy integration of new features, scalability of the database, and compatibility with external systems. The backend ensures secure data storage, fast retrieval, and reliable system performance, while fallback mechanisms enhance system robustness (Wolfert et al., 2017).

1.5.3 Educational and Social Objectives

The platform is designed to promote sustainability and social development within the agricultural sector. It encourages eco-friendly farming practices by supporting millet cultivation, which requires fewer resources and is resilient to climate change (NITI Aayog, 2018).

Furthermore, the platform empowers farmers by providing access to market insights and digital tools, enabling informed decision-making and increased productivity. It also contributes to rural economic development by improving market connectivity and reducing inefficiencies in the value chain.

The interactive nature of the platform promotes self-learning and user engagement, allowing users to explore and understand various aspects of the millet ecosystem effectively.

1.5.4 Accessibility Objective

The platform aims to ensure inclusivity and accessibility for all users, regardless of their technical expertise. It features a simple user interface, mobile-friendly design, and low data consumption to accommodate users in rural areas. Future enhancements such as multilingual support and voice-based interaction will further improve accessibility and usability across diverse user groups (OECD, 2020).

1.6 Sustainable Development Goals (SDGs)

The Millets Value Chain Platform aligns with several United Nations Sustainable Development Goals (SDGs), contributing to broader global objectives such as food security, sustainability, and economic growth.

Figure 1.1 Sustainable development goals

SDG 2: Zero Hunger

The platform promotes millet production and consumption, enhancing food security and supporting sustainable agriculture. Millets are highly nutritious and contribute to diversified and resilient food systems (FAO, 2023).

SDG 12: Responsible Consumption and Production

By encouraging the consumption of environmentally sustainable crops such as millets, the platform supports responsible resource utilization and reduces dependency on water-intensive crops like rice and wheat (Government of India, 2023).

SDG 13: Climate Action

Millets are climate-resilient crops requiring fewer natural resources. The platform promotes sustainable agricultural practices that reduce environmental impact and support climate adaptation (FAO, 2023).

SDG 8: Decent Work and Economic Growth

The platform enhances farmers' income, supports rural entrepreneurship, and improves market access, contributing to economic development and employment opportunities (Adekunle et al., 2018).

SDG 10: Reduced Inequalities

By providing accessible digital tools and market information, the platform reduces inequalities and supports small-scale farmers, particularly in rural areas, thereby bridging the digital divide (Tripathi et al., 2022).

1.7 Overview of Project Report

The project report will be in a structured and organized format in order to understand the Millets Value Chain Platform, its conceptual base, technical implementation, and the impact it gives comprehensively. Every chapter is aimed to cover some particular issue of the project starting with the background and moving through the research, development, analysis and concluding proposals.

Chapter 1 – Introduction

This chapter defines the Millets Value Chain Platform and sets the scene of the project. It gives the history of the millet agriculture, the importance of millets in sustainable farming, the necessity of digital transformation in the millet ecosystem, project facts, current technologies, future undertaking, adherence to Sustainable Development Goals (SDGs), and the general organization of the report.

Chapter 2 – Literature Review

This chapter provides the review of the research outcomes, agricultural platforms, and academic literature concerning digital agriculture, millet value chains, supply chain integration, and user-centric agricultural systems.

It also outlines the advantages and disadvantages of current solutions and outlines the gap in the study that the proposed platform would contribute to bridging.

Chapter 3 – Methodology

In this chapter, the approach to be applied in the project development is unveiled. It explains the Hybrid Agile model, the analysis of requirements, the prototyping, system design, the front end and the back-end development, the testing, integration and refining as well as the ultimate implementation process. It also helps to choose the methodology in this project.

Chapter 4 – Project Management

The focus of this chapter is on planning and management of the project. It includes the project time line, development phases, to track the milestones, identification and mitigation strategies of risk, budget planning. It shows the structuring and the way the project has been implemented in an efficient manner with the academic and technical limitations.

Chapter 5- Analysis and Design

This chapter gives the technical design and the system design of the platform. It has non-functional and functional requirement, content requirement, system architecture, block diagram, flowchart, modular unit design, and important design decision taken during system development.

Chapter 6- Software Development Tools

In this chapter, software, framework, and technologies used to implement the project are described. It contains HTML, CSS, JavaScript, React.js, Node.js, Express.js, MySQL, JSON, JWT authentication, file handling tools, development environment tools, and debugging tools, and the usage and appreciation of each technology.

Chapter 7 Social, Legal, Ethical, Sustainability, and Safety Aspects

The chapter also evaluates the big picture impacts of the overall project implementation other than those of technical systems. It talks about the social importance of the platform, one that empowers farmers, brings it closer to the farmers, its legal aspects of data and content about users, moral obligations in online agriculture, sustainability, and risk mitigation of its systems.

Chapter 8 – Conclusion

The chapter is an overview of the overall work undertaken in the project. It describes the great achievements, the most common outcomes, and limitations of the project, and recommendation on how to enhance the platform. It also speculates on how the project will attain the intended purpose and whether the purpose can be applicable in a real-life scenario.

References:

It is here that all the research papers, journal articles, reports, government documents and the internet resources, which were consulted in the course of the project, are located. All the supporting sources are referred to by proper academic citation standards.

Appendix:

The supporting materials of the project such as details of the publication, similarity report, a link to the GitHub repository, a link to live demo, and some screenshots of the system, and other evidence that were utilized in documentation of the project are provided in the appendix.

Chapter 2

Literature Review

The literature review gives the scholarly base of the Millets Value Chain Platform by studying the available research studies, agri-digital systems, supply chain models and technological solutions used in the agri-food industry. It assists in realizing the advantages and weaknesses of existing systems and locates research gaps that prove the necessity of the offered platform. The review is organized into thematic sections like digital transformation, millet value chain issues, system constraints, and user-centric design strategies.

2.1 Digital Revolution in Agricultural Value Chains in India

In developing countries, agricultural value chains are usually described as being fragmented, uncoordinated and inefficient in a manner that impacts both productivity and income of farmers. According to Trienekens (2011), a lack of integration between the stakeholders and inadequate infrastructure leads to low transparency and low economic gains to farmers. The movement towards the digitalization of agricultural systems has increased with the development of digital technologies.

Digital platforms provide access to the real-time market, enhance supply chain visibility, and direct contact between farmers and buyers. Sharma and Singh (2021) note that the use of digital agricultural platforms in India results in efficiency due to the minimized number of intermediaries and the availability of more efficient prices to farmers. Likewise, Tripathi et al. (2022) underscore the significance of digital marketplaces that are farmer-centric in enhancing accessibility and involvement in the agricultural trade.

The efforts by the government have also been a major factor in supporting digital agriculture. The Government of India (2023) has come up with different steps by the National Year of Millets to promote digital infrastructure and assist farmers. OECD (2020) also mentions that digital technologies are used to enhance coordination and transparency of food supply chains all over the world.

Nevertheless, as much as this has been achieved, the current platforms do not offer end-to-end integration and are still constrained in terms of handling particular crops like millets.

2.2 Importance of Millet Value Chain Integration

Millets are well known in terms of nutritional values and climate resilience but their value chain has not been developed as the other staple crops. Dayakar Rao et al. (2017) emphasize that millets are good sources of essential nutrients that play a major role in food security and health. FAO (2023) also highlights that millets are important in sustainable food systems because of their minimal water needs and ability to cope with adverse climatic conditions. In spite of these benefits, there are a number of structural issues affecting the millet ecosystem. Lack of organized supply chains, ineffective market access, and inadequate processing infrastructure are some of the issues that are identified by Shanmuka and Shireesha (2018). Adekunle et al. (2018) also note that small millet agribusinesses have difficulty in growing because of poor coordination among the stakeholders and lack of institutional support. Banu et al. (2019) study the millet value chain in the Indian context and point to the inefficiencies of the production, distribution, and marketing processes. On the same note, Sangappa et al. (2020) note that Farmer Producer Organizations (FPOs) can enhance the millet value chain through better collective marketing and decreasing reliance on intermediaries. Nevertheless, they are yet to be digitally integrated and real-time coordinated. The adoption of millet products is also dependent on the consumer perception. According to Vattekkad et al. (2020), consumer demand is low due to low awareness and availability of value-added millet products. Muskan et al. (2022) also mention that the global millet markets are characterized by insufficient promotion, irregular supply, and ineffective distribution channels. Thus, the millet value chain needs to be integrated using a digital platform to increase the level of coordination, transparency, and encourage the use of millet.

This explains why the proposed system should include characteristics like farmer dashboards, supply chains tracking, marketplace integration and educational modules.

2.3 Weaknesses of the Current Agricultural Digital Solutions

The current agricultural digital systems have already helped in enhancing market access and availability of information, but they continue to have a number of shortcomings.

According to Jayalath et al. (2021), as much as digitalization of agri-food supply chains enhances governance and efficiency, most of the systems are not well integrated at various parts of the supply chain. Another significant issue in agricultural systems is traceability. Charlebois et al. (2021) discuss digital traceability in food supply chains and emphasize the fact that despite the adoption of the state-of-the-art tracking mechanisms by developed countries, the implementation issues continue to be a challenge in many developing regions. Equally, Ahmed et al. (2021) find that the agrifood supply chain traceability systems are generally not standardized and consistent, which influences reliability. Moreover, technologies like blockchain have been suggested to enhance transparency. Kamilaris et al. (2019) explain the possibilities of blockchain in agriculture in terms of guaranteeing safe and transparent transactions, but also mention that there are some issues with scalability and adoption. Wolfert et al. (2017) also emphasize that big data technologies can improve smart farming; however, their usability relies on the appropriate integration and management of data. The other urgent problem is a lack of crop specific focus. The majority of available platforms are geared towards general agricultural products and lack specialization in supporting millets. Moreover, usability issues exist, especially to small-scale farmers due to their limited digital literacy. These shortcomings suggest that there must be an integrated, easy-to-use, and millet platform.

2.4 Multimedia and Data Visualization in Agriculture

Multimedia tools and data visualization are important in enhancing understanding and decision-making in agriculture. According to Wolfert et al. (2017), data-driven strategies in smart farming assist farmers in studying the trends, forecasting the results, and making reasonable decisions. Presentation of complex data in its visual form makes information easier to understand and improves the interaction with the user. Another method of knowledge dissemination is multimedia (videos, interactive modules). FAO (2023) emphasizes that educational resources and awareness programs are crucial to fostering sustainable agricultural practices. Digital platforms can enhance user-experience and promote active engagement, using data visualization and multimedia. The implementation of these concepts in this project includes interactive dashboards, graphical tracking of the supply chain, and educational materials concerning the millet cultivation and advantages. This practice takes the conventional data presentation into an exciting and informative experience.

2.5 Interactive Learning and Gamification of Agricultural Platforms

Gamification is just one of a growing number of interactive learning methods utilized to make users more engaged within digital platforms. Despite the fact that it is not adopted broadly in the agricultural systems, studies conducted in the digital learning systems demonstrate that interactive capabilities enhance user engagement, retention and motivation of knowledge. Agricultural platforms can promote farmers to use more appropriate practices and actively use the platform by including features like progress monitoring, rewards, and interactive content. Such characteristics can be especially helpful to create awareness about the benefits of millet and the encouragement of sustainable farming methods.

2.6 Accessibility to Websites and User-Friendly Farm interfaces

The success of digital agricultural platforms heavily relies on accessibility and usability. According to Sharma and Singh (2021), a significant number of platforms do not cater to the needs of rural users, especially those with low digital literacy. The typical pitfalls are intricate navigation, data intensive, and mobile unfriendliness. Tripathi et al. (2022) further point out that it is crucial to design with farmers to provide adoption and efficient use. The platforms should be straightforward, mobile, and work with devices of different internet connectivity. The proposed platform will overcome these challenges since it offers a clean and intuitive interface, mobile-friendly design, and efficient data processing with lightweight technologies. Accessibility and inclusivity can be further improved by future improvements like multilingual support and voice-based assistance.

2.7 Summary of Literature Review

It is evident in the literature review that although a lot has been achieved in the area of digital agriculture, various gaps still remain in the millet value chains. The current systems offer biased options like access to markets and advisory services, and they lack a full integration, real-time monitoring and crop-centric attention. Digital transformation, supply chain integration, traceability and user-centric design have been highlighted in research studies as necessary to enhance agricultural systems. Nonetheless, shortcomings like disjointed information, poor farmer participation and awareness still persist.

The Millets Value Chain Platform fills these loopholes by providing a centralized online platform that connects farmers, buyers, and stakeholders. It offers supply chain management, shopping capabilities, learning tools and interaction in one platform. The system will be used to increase transparency, increase farmer income, and sustainability of agricultural practices by integrating technological innovation with user-centered design.

Chapter 3

Methodology

The design of the Millets Value Chain Platform was done through a designed hybrid approach which integrates the flexibility of Agile practices with the predictability and methodical character of conventional software development models. This method not only guaranteed a continuous feedback process, iterative refinement and polishing during the development process but also corresponded to academic needs and systematic review cycles. This chapter describes the methodology used, the lifecycle steps taken in the development and justification of the Hybrid Agile model choice. The methodology has been developed in compliance with the typical software engineering standards to provide technical strength with usability.

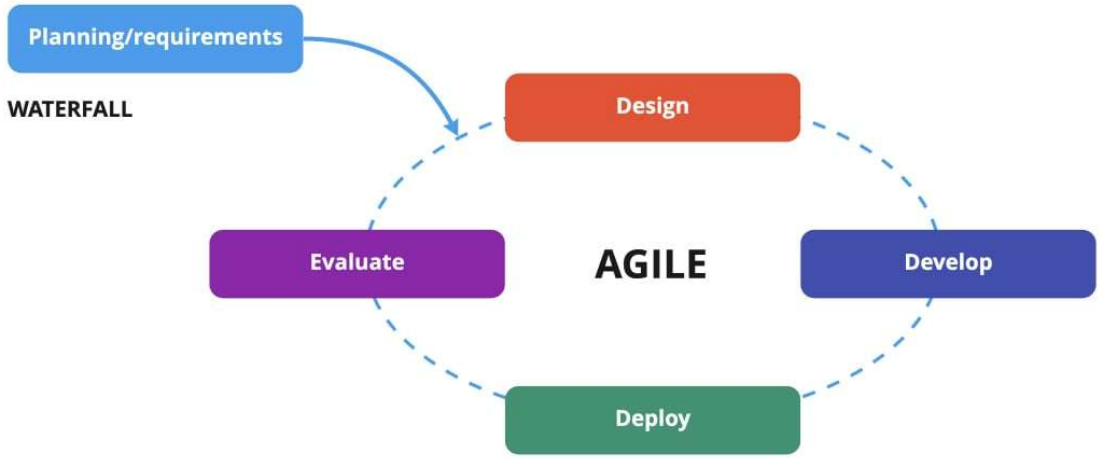

Figure 3.1 Hybrid Agile Methodology

3.1 Project Development Methodology

The project is developed in Hybrid Agile Approach that merges iterative development method with a well-organized implementation in phases. This methodology enabled the system to be created in bits with appraisal and refinements being done after every phase. Concurrently, it made sure that the project was kept in line with academic milestones and documentation requirements. The methodology offered flexibility to include feedback obtained during reviews and permitted one to make changes without interfering with the general structure of the system.

Meanwhile, it also had a development path that was well clear, which ensured that the project had an organized and time bound development. This flexibility and structure balance allowed the Hybrid Agile model to be appropriate when developing this platform.

3.2 Project Development Lifecycle Stages

The project was created in a few steps which resulted in the gradual optimization and stabilization of the system. These phases resulted in the platform being planned, developed, verified and accepted systematically.

3.2.1 Phase 1: Preliminary Research and Requirement Analysis

The first stage was on knowledge of the problem area and the needs of the system. An in-depth analysis of the current agricultural platforms like eNAM, AgriStack, and other private agri-tech solutions was performed. In this analysis, there were various limitations such as a lack of integration, limited interactivity, and non-millet-specific digital solutions, which are also noted in the literature on digital platforms in agriculture (Sharma and Singh, 2021).

On the basis of this analysis, it was revealed that the main gaps were the necessity to have a single millet value chain platform, better connectivity of farmers-to-market, and enhanced transparency in supply chain operations. A dedicated marketplace and tracking mechanisms, as well as interactive dashboards, were also needed to facilitate the system.

This was followed by gathering of requirements to establish the system scope. These functional requirements consisted of farmer, marketplace, and supply chain tracking modules. Non-functional requirements were related to performance, scalability, and usability.

This was followed by requirement gathering to outline the scope of the system. Functional requirements referred to farmer modules, marketplace operations and supply chain tracking modules. Non-functional requirements were performance, scalability and usability requirements. Also, data requirements like information about millet, farmer records and transaction data were established. The accessibility needs provided the platform with simplicity, mobile-friendliness, and an ability to be used by users with different levels of technical expertise. This stage eventually defined the boundary of the system and gave an idea where to go with development.

3.2.2 Phase 2: Prototype Design and validation

To have a visualization of the system prior to full-scale development, a prototype was developed. This step aided in the designation of the layout, structure, and user interaction flow of the platform. The prototype contained major interfaces, which include home page, farmer dashboard, marketplace, supply chain tracking module, and product listing pages.

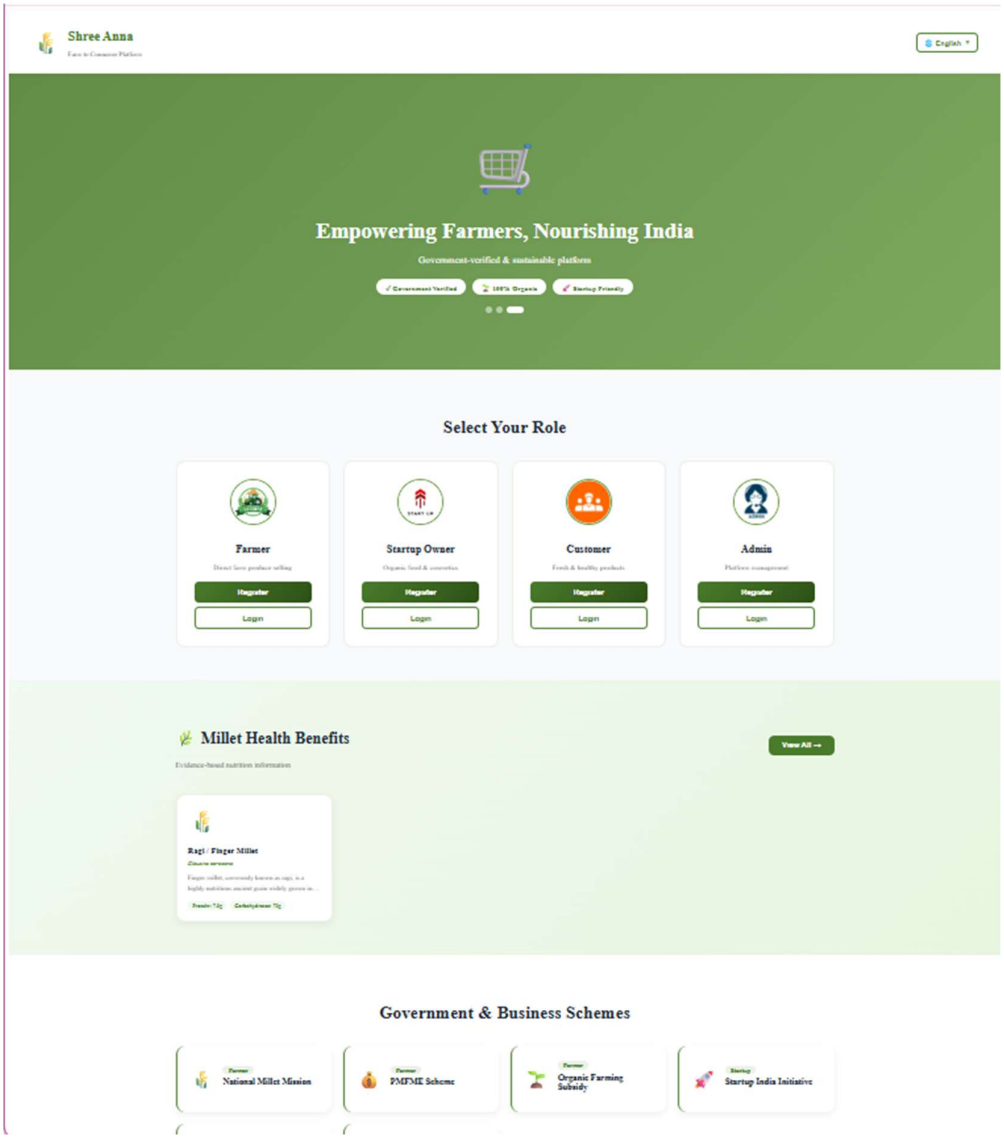

Figure 3.2 Design Prototype – Homepage

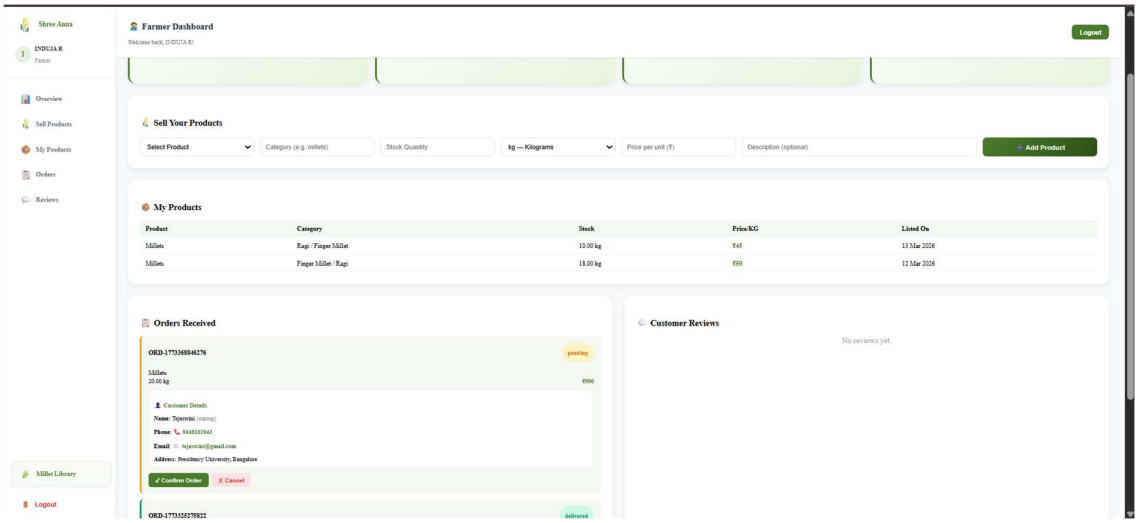

Figure 3.3 Farmer Dashboard Layout

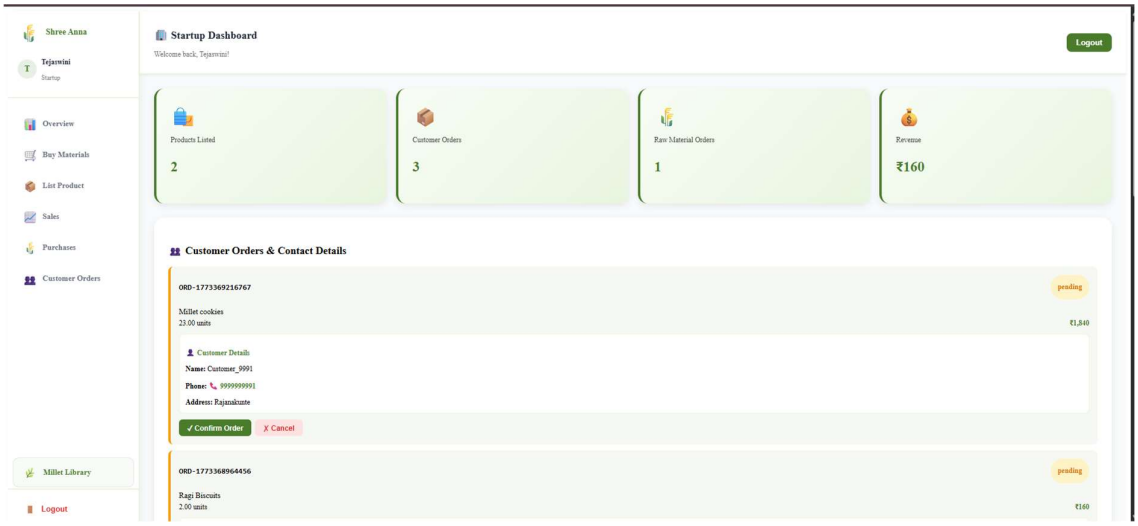

Figure 3.4 Startup Dashboard Layout

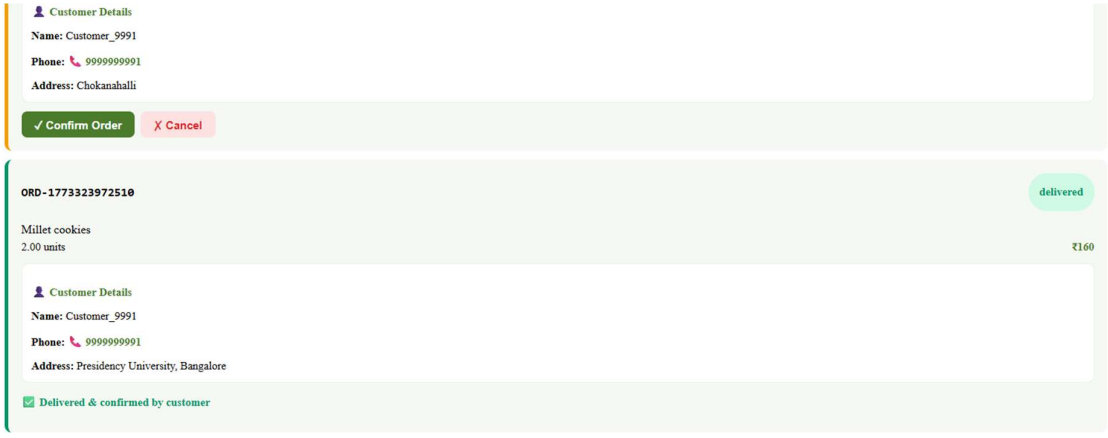

Figure 3.5 Supply Chain Tracking Interface

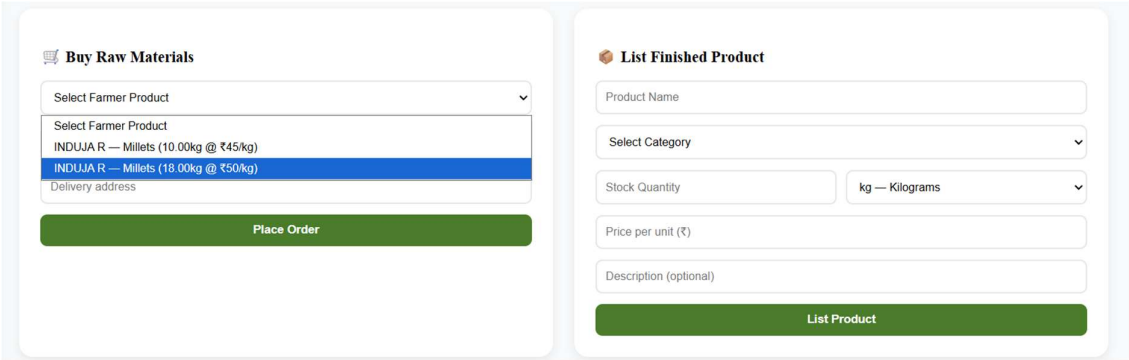

Figure 3.6 Product Listing Layout

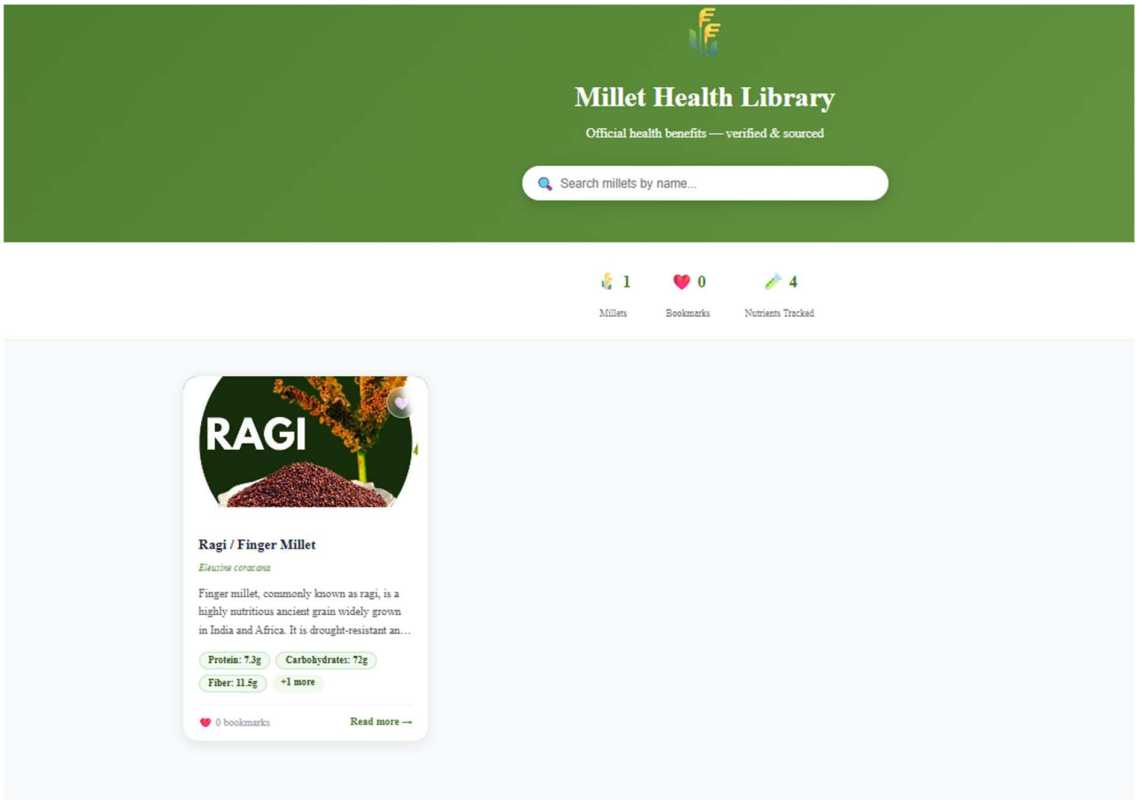

Figure 3.7 Millet Library Layout

The prototype was introduced in the first review, and the feedback was obtained on how to improve the user interface design, make information more balanced, flow more natural, and add interactive dashboards. Depending on this feedback, the design was narrowed down to enhance usability and overall user experience and then it moved to the development stage.

3.2.3 Phase 3: Backend Development using Node.js and MySQL

The third step involved the creation of the main functionality of the platform by implementing the backend. MySQL was used to create a structured database schema to store data on farmers, products, supply chain phases, transactions and users. This guaranteed a well-organized and effective data management.

The back end was built on Node.js and Express.js to develop RESTful APIs. These APIs dealt with different functions including user authentication, product management, order handling and data retrieval. The main data exchange format between the frontend and the backend was JSON which provided an efficient means of communication.

The choice of Node.js was based on the non-blocking design, scalability, and the ability to handle multiple simultaneous requests cost-effectively. MySQL has been selected due to its high relational data support, structured storage, and query processing. Other mechanisms like fallback handling were introduced to enhance the reliability of the system and facilitated smooth working of the system under various conditions.

3.2.4 Phase 4: Frontend Development using HTML, CSS, JavaScript, and React.js

During this stage, the platform user interface was created and made to be interactive and user friendly. The frontend was structured with HTML, styled with CSS, made interactive with JavaScript, and dynamic and reusable with React.js.

It was developed with the aim of developing responsive layouts that work effectively with various devices like desktops, tablets, and mobile phones. To ensure consistency throughout the platform reusable components like dashboards, forms and data display cards were adopted. The use of APIs allowed the real-time update of data and the seamless communication between the interface and back-end systems.

React.js was very instrumental in state management of applications and in displaying dynamic content in an effective manner. Its component-based design enhanced maintainability and scalability and its virtual DOM provided faster updates and enhanced performance. Altogether, this stage made sure that the platform was an enjoyable user experience.

3.2.5 Phase 5: Integration, Testing and Refinement

After developing the frontend and backend components, they were combined into a whole system. This stage included mapping APIs to user interface, database operations to application logic and real-time flow of data across modules (dashboards and supply chain tracking).

Extensive testing was done to make the systems reliable and performant. These encompassed unit testing of each module, API testing with integration, testing responsiveness in various devices, and testing cross-browser compatibility. Also, user-interaction testing and simple load testing were conducted to test the system behaviour under various conditions.

At this stage, mistakes and inconsistencies were detected and rectified. The system was optimized to enhance performance, stability, and usability making sure that it met the desired functional and technical requirements.

3.2.6 Phase 6: Finalization and Documentation

The last stage was aimed at finalizing the system and making it ready to be submitted and evaluated. All modules and features were completed and the user interface was streamlined to make it easier to use and to access. Particular care was taken to make sure that the system is user friendly particularly to non-technical users.

To explain the system design, implementation and functionalities in detail documentation was done. Presentation materials were also created to demonstrate the working of the platform. Last testing and verifying was conducted to make sure the system works properly and that it meets all the outlined requirements.

This step made sure that both the technical and academic part of the project were fully complete and thus ready to be evaluated finally.

3.3 Reason for Adopting Hybrid Agile Methodology

The Hybrid Agile methodology was chosen because of the necessity to find the balance between flexibility and structure. Waterfall would not enable iterative enhancements, whereas the completely Agile style might not fit well with the academic schedule and documentation needs.

The hybrid model also created the opportunity of an iterative development based on Agile practices and made it possible to have constant feedback and improvements. Meanwhile, it was still structured in a manner that resembled the Waterfall model, with documentation and tracking of milestones being taken care of. This mix allowed developing it efficiently and at the same time, being consistent and of high quality across the project.

3.4 Summary of Methodology

The methodology adopted made sure that the system was designed in a systematic and organized way whilst flexible to allow on-going improvement. The feedback on the various levels helped in the improvement of the quality of the system and errors were detected and corrected at an early stage of the development process.

The methodology, through a combination of technical efficiency and user-centric design, guaranteed the end system to be functional, technically effective and to the expectations of the academic.

Chapter 4

Project Management

Project management was instrumental in ensuring that the development of the Millets Value Chain Platform was done efficiently in all levels such as planning, design, implementation, and testing. This chapter includes the management practices, timeline of development, risk assessment and cost planning that led to successful project delivery.

The project was undertaken in a structured and organized process in accordance with academic milestones and review phases as well as flexibility being influenced by Agile development principles. This practice has made a continuous improvement possible with the help of the iterative feedback and guaranteed the progress in time. The project schedule, risk analysis, and cost estimation are explained in the following sections.

4.1 Project Timeline

The project development was split into several phases, each of which corresponded to the review cycles of the university. Each stage involved certain activities, products, and the results of evaluation. This systematic advancement guaranteed a systematic growth and constant improvement on the lines of feedback.

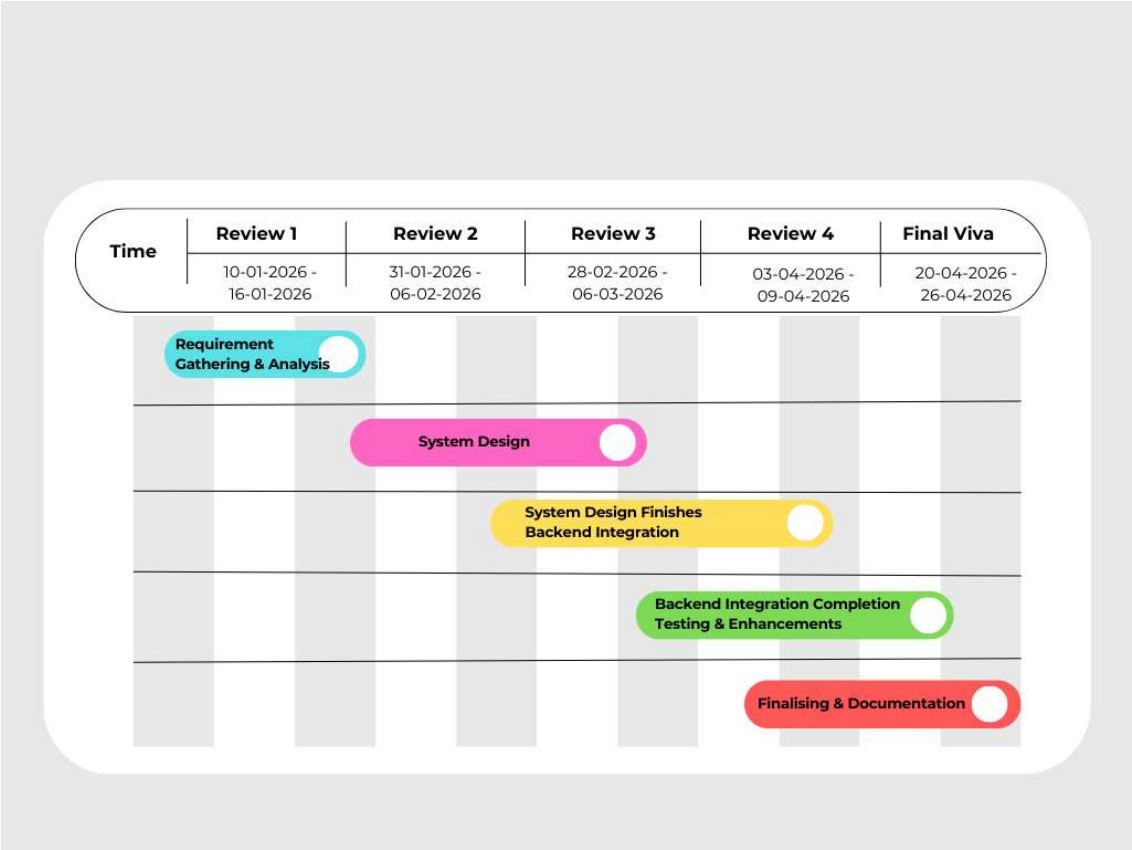

Figure 4.1 Project Timeline

Phase I- Research and Requirement Analysis (Before Review 1)

Activities:

- Conducted literature review on agricultural systems and millet value chains
- Studied existing agri-tech platforms such as eNAM and AgriStack
- Identified gaps in millet-specific digital solutions
- Defined functional and non-functional system requirements
- Finalized project scope and objectives
- Designed initial system architecture

Deliverables:

- Requirement Specification Document
- Initial Concept Document
- Draft of technical approach

Outcome:

Review 1 confirmed the project concept and gave the project general direction a go ahead.

Phase II- Prototype Design and UI Layouts (Prior to Review 2)

Activities:

- Designed initial website wireframes
- Developed the basic homepage layout
- Implemented UI components using HTML, CSS, and React
- Defined navigation structure and page flow
- Created a functional prototype

Deliverables:

- Initial working version of the platform
- UI/UX design preview

Outcome:

The feedback made in review 2 showed the need to make visuals better, provide additional sections, and make it more interactive. The following step incorporated these.

Phase III Core Implementation and Backend Development Before Review 3)

Activities:

- Designed MySQL database schema including:
  - Farmers
  - Products
  - Supply chain data
  - Transactions
- Developed backend server using Node.js
- Implemented RESTful API routes
- Integrated major modules such as farmer dashboard, marketplace, and tracking system
- Developed core functional pages

Deliverables:

- Integrated frontend and backend system
- Functional modules ready for testing
- Draft version of the project report

Outcome:

Review 3 verified a solid system base and recommended additional polishing and the addition of features.

Phase IV - Integration, Test and Refining (Before Review 4)

Activities:

- Incorporated all modules (dashboard, marketplace, tracking system)
- Linked frontend to back-end APIs
- Performed cross-device responsiveness testing
- Identified and fixed software bugs
- Population in the completed data in the database

Deliverables:

- Fully functional platform
- Operational modules
- Testing documentation

Outcome:

Review 4 certified preparedness to final assessment.

Phase V - Finalization and Submission (Prior to Final Viva)

Activities:

- Final project report
- Creation of presentation slides
- Final system validation

Deliverables:

- Final report
- Final presentation
- Complete working system
- Demonstration-ready platform

4.2 Risk Analysis

To provide seamless execution of the project, risk analysis was necessary. Risks were defined and identified at an early stage and classified into scope risks, operational risks, and technical risks. Proper mitigation measures were taken to ensure they had minimal effects

4.2.1 Technical Risks

Risk 1: Reliance on External Data Sources

Reliance on third-party services or external APIs can also cause failure in the system in case these services are unavailable.

Mitigation:

- Implemented fallback mechanisms
- Local storage of important data by use of JSON and MySQL

Risk 2: Performance Problems with Large Data

Processing vast amount of agricultural and supply chain data can influence the performance of system.

Mitigation:

- Structured JSON data storage optimization
- Less unwanted data transfer
- Created an effective UI implementation in React

Risk 3: Cross-Browser Compatibility Problem

There can be inconsistencies in the way UIs behave across browsers.

Mitigation:

- Verified with various browsers (Chrome, Edge, Firefox)

Used standardized web technologies.

4.2.2 Operational Risks

Risk 4: Insufficient Development Period Before Reviews

The time between review stages can be a limitation to feature development being short.

Mitigation:

- Prioritized core functionalities
- Adhered to iterative development process
- Well differentiated development activities

Risk 5: Data collection delay

Gathering data on millet related data in a structured manner can be time consuming.

Mitigation:

- Applied pre-designed templates to enter data
- Conducted parallel data collection in the course of development

4.2.3 Scope-Related Risks

Risk 6: Scope creep (Introduction of Unnecessary Features)

The process of adding more features can slow down a project.

Mitigation:

- Clear project delimitations
- Put off other features to future scope

Risk 7: Continuous UI Modifications

Consistency may be impacted by frequent design changes.

Mitigation:

- Set up a definite design framework early
- Unspoilt reusable UI components in React

4.3 Project Budget

Millets Value Chain Platform was created with the help of open-source tools and free-at-will resources, and it is, therefore, very cost-effective.

4.3.1 Software and Development Tools

Table 4.1 Software and Development Tools

Item	Tool Used	Cost
Code Editor	Visual Studio Code	Free
Backend Framework	Node.js + Express	Free
Database System	MySQL Community Edition	Free
Frontend Tools	React js, HTML, CSS, JavaScript	Free
Version Control	Git + GitHub	Free
Testing Tools	Browser Developer Tools	Free
Deployment	Netlify / Vercel (free tier)	Free

4.3.2 Content and Multimedia Assets

Table 4.2 Content and Multimedia Used

Requirement	Source	Cost
Agricultural Data	Public datasets / manual entry	Free
Images	Royalty-free / self-created	Free
Videos	Embedded online content	Free
Informational Content	Manually curated	Free
Icons & Fonts	Open-source libraries	Free

4.3.3 Hardware and Infrastructure

Table 4.3 Hardware and Infrastructure Used

Item	Description	Cost
Development System	Personal laptops	No additional cost
Internet	Institutional / personal connection	No additional cost

Total Cost of Project:

₹ 0 (Zero Budget)

The project proves that it is possible to create a scalable and effective digital agricultural platform without spending a huge sum on its development due to the use of open-source technologies and the efficient use of resources.

Chapter 5

Analysis And Design

Analysis and Design phase is the phase of systematic description and structure of the technical structure of the MVCP. The system is subdivided into logical components in this chapter to give a clear picture of how the system works. It describes actual visible functions that can be performed, as well as the processes that take place internally in the system.

The design is presented in form of architectural features like block diagrams, flowcharts and modular structures. Such representations aid in the comprehension of interactions between various components. The main goal of this stage is to make sure that the system is scalable, efficient, and easy to use by stakeholders that include farmers, buyers and administrators of the millet ecosystem.

5.1 Requirements

The requirements phase involves determining what the system needs to do, and under what circumstances it needs to run. These specifications were based on various sources such as research analysis, user expectations, project guidelines, and the limitations that were experienced in the available agricultural platforms.

The requirements are divided into three groups, namely, functional requirements, non-functional requirements and content requirements in order to bring about clarity and structure.

5.1.1 Functional Requirements

Functional requirements specify the fundamental functions that the system should carry out. These needs guarantee that the platform facilitates all the important activities associated with the millet value chain.

The system has a farmer registration and management module which enables farmers to make accounts, secure logins and manage the account. All the data of farmers is securely stored and maintained in the system.

It also has a product listing option, which allows farmers to post millet products like ragi, jowar, and bajra besides price, quantity and availability. This makes products well displayed to the potential buyers.

The platform will have the supply chain tracking functionality that will be used to track the movement of millet products through supply chain stages such as production, storage, transportation, and final sale. This enhances transparency across the supply chain.

A buyer and marketplace module will be offered to enable customers to navigate through the products, see the detailed description, and communicate with sellers. This forms a direct relationship between consumers and producers.

The other important feature is transaction management which deals with the placement, processing and tracking of orders. It guarantees a seamless process of buying and selling products on the platform.

The system also has a dashboard module which shows real-time data like product listing, transaction details and supply chain updates. This will allow the users to make sound decisions utilizing the available data.

Lastly, the platform has an adaptive interface that can accommodate various gadgets like desktops and mobile systems. React.js and standard CSS will be used to provide a dynamic and versatile experience to users with different screen sizes.

5.1.2 Non-Functional Requirements

Non-functional requirements specify the performance of the system and the quality characteristics that have to be observed throughout the system operation.

The site will be capable of high performance so that page loads will be within a reasonable range of 3-5 seconds. This is done by an efficient data handling mechanism and optimized API calls.

Scalability is also an important factor to consider, which enables the system to support growing numbers of users, products and transactions on the platform as it expands. The architecture will be able to support the expansion in the future without significant structural modifications.

Stability in behaviour of the systems and inclusion of fallback mechanisms to address unexpected failures ensures reliability. This ensures that performance remains constant even in case of different conditions.

The interface is simplified and intuitive, thus highlighting the concept of usability, allowing the system to be used by users of various technical knowledge, especially rural farmers.

The platform is designed to be mobile-friendly and easy to navigate, which also increases accessibility since users can use the platform regardless of their devices and the environment.

Modular code design with React components and Node.js APIs also ensures maintainability and allows more convenient updating, debugging, and extending the system in the long run.

5.1.3 Content Requirements

Content requirements state what kind of data the system has to operate with and manage efficiently.

The site keeps a record of the details of the farmers, such as the profile and registration details. It also maintains millet product information like type, price and availability.

The information of the supply chain stages is kept to monitor the flow of products at various stages. To track the buying and selling processes in the system, the records of transactions are kept.

Moreover, the system will produce dashboard analytics data to give information on system performance and user actions. It is also necessary that all information is accurate, properly organized and manageable to ensure a smooth running of the system.

5.2 Block Diagram

The block diagram provides a high-level overview of the system architecture. It demonstrates the connection between various parts and their relationship to guarantee the efficient operation of the platform.

The diagram contains the user interface created with React.js, the backend system created with Node.js and MySQL database to store the data, supply chain logic to trace the processes and communication in the form of a JSON to exchange the data.

These are interrelated components to create a complete system where information passes smoothly without any problem between frontend, backend, and database layers.

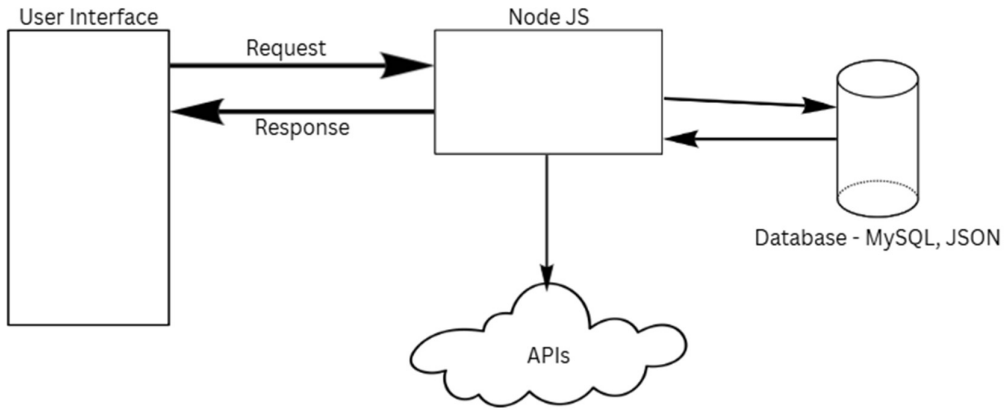

Figure 5.1 Block Diagram

5.2.1 System Architecture Description

The system architecture is made up of several interrelated components that collaborate to provide the functionality of the platform.

The frontend layer is the user interface which is written in HTML, CSS, JavaScript and React.js. It gives access to the users to dashboards, product listings, and supply chain information and interacts with the backend via API calls.

The frontend sends requests to the application server which is developed with Node.js. It handles business logic, interacts with the database, and generates appropriate responses.

MySQL database is the database layer that stores structured data about the users, products, transactions and supply chain phases.

As the data exchange format, JSON is employed, which allows communicating effectively between the frontend and the backend elements.

The supply chain logic module is in charge of products tracking at various phases and maintaining transparency and traceability.

The flow of data in the system is organized and user actions are sent to the backend by the frontend, processed, accessed by the database and sent back to the frontend to be displayed.

5.3 Flow Chart

The flowchart illustrates the working process of the system. It graphically illustrates the way the user actions are handled and the flow of data between the various layers of the application.

It assists in knowing the flow of operations, beginning with user interaction to back-end processing and end output.

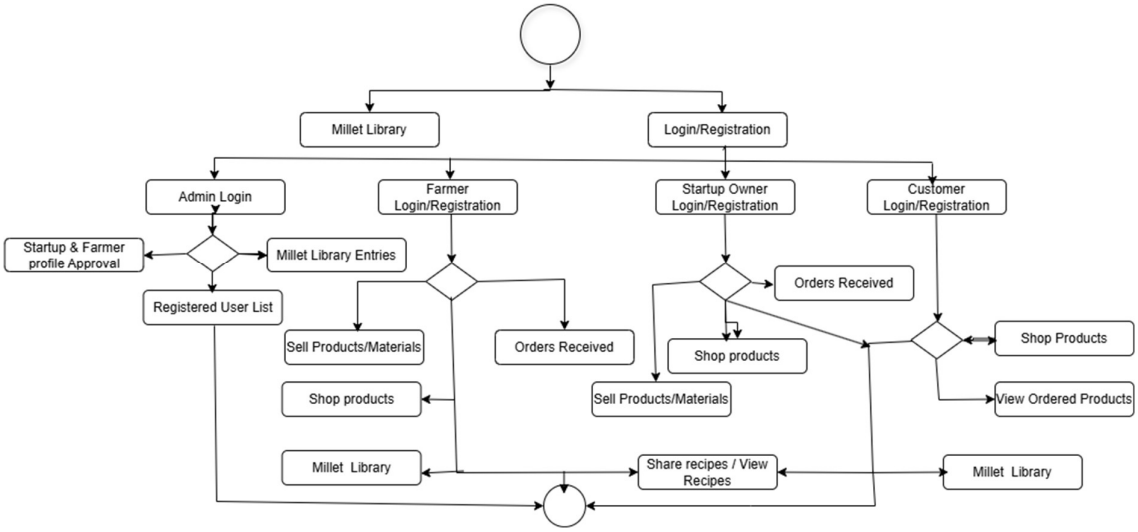

Figure 5.2 Flow Chart

5.3.1 System Workflow Explanation

The system starts at the user interaction point, where a user enters the platform and chooses an operation, whether it is adding a product, exploring the market-place, or monitoring the supply chain information.

After choosing an action, the frontend makes a request to the server of Node.js via API calls.

The backend, based on the business logic, processes the request by retrieving the appropriate data in MySQL database. This can be in the form of information about farmers, products or transactions.

Once processed, the server responds with a response containing the necessary data in the form of a JSON.

Lastly, the frontend is dynamically updated with React.js to show the information requested and has a responsive and fluid user experience.

5.4 Device Compatibility

The software is web based, and therefore, it is compatible with a variety of devices.

Access to clients is mainly via the laptops and desktop computers, but mobile devices can also be used on the platform. The responsive design will make sure that the interface is responsive towards the various screen sizes and resolutions.

The application is compatible on the server side with any system with Node.js support, both Windows and Linux. This is flexible and can be deployed and scaled easily.

5.5 Designing Units

The platform is split into smaller modular units to enhance its efficiency in development and organization of the system. It is a modular approach, which is in line with Agile development practices and enables various components to be developed and tested separately.

5.5.1 Unit Breakdown

Table 5.1 Unit Breakdown

Unit	Technology	Function
Farmer Module	React + Node.js	Farmer registration and management
Product Module	MySQL + Node.js	Millet product listing
Marketplace	React + API	Buyer interaction
Supply Chain	Node.js Logic	Product tracking
Transactions	MySQL	Order and payment tracking
Backend APIs	Node.js + Express	Data routing
Database Layer	MySQL	Data storage
UI Layer	React + CSS	Frontend display

5.5.2 Modularity Benefits

A modular design of the system has a number of benefits. It makes upgrading or the addition of features to the system simple without compromising the whole system. Scalability is facilitated by the structure, which means that the platform can expand with the growth of the user demand.

It also allows debugging to be done faster since problems can be noted and fixed in individual modules as opposed to the entire system. The reusable elements, especially the frontend elements, minimize the redundancy and enhance the efficiency of the development process.

On the whole, the modular approach guarantees flexibility, maintainability, and sustainability of the platform in the long run.

Chapter 6

Software Development Tools

The evolution of the Shree Anna Millet Supply Chain Platform was a chance to select technologies carefully at every stage, not only due to their popularity, but also to their appropriateness to create a scalable, efficient, and user-friendly system. The platform will cater to a variety of stakeholders, such as farmers in the rural environment, owners of startups handling orders, and administrators managing the ecosystem. Thus, the selected tools guarantee ease, functionality, and sustainability.

6.1 HTML (HyperText Markup Language)

All the pages of the platform are based on HTML. It is employed to organize the registration forms, product listing, dashboards and order interfaces. Starting with farmer onboarding forms to admin dashboards, all visual elements start with HTML. The use of HTML facilitates universal compatibility and rapid loading which is more so crucial to users in rural settings who may be using low bandwidth network or have older machines. The simplicity of it gives it a stable foundation on which to incorporate other technologies.

6.2 CSS (CSS inline styles through React)

Rather than relying on a different CSS framework, inline CSS was used in React components to style. The method is able to make each part stand-alone and prevent style clashes between parts of the application. The entire platform was covered with a homogeneous color scheme using the hues of green to indicate its agricultural theme. The use of common style patterns was replicated on different components like buttons, cards and tables, which guaranteed visual consistency and ease of maintenance.

6.3 JavaScript (Frontend Logic)

All the interactive features in the platform are undertaken by JavaScript. It manages dynamic actions like form submissions, API calls, real-time updates, modal interactions, and notifications without page reloads.

This allows users to make orders, status updates and reviews in a smooth and responsive way. JavaScript has been used to make the application more interactive and quicker hence improving the user experience.

6.4 React.js (Frontend Library)

The front of the platform is fully constructed with the help of React.js, which enables the application to be designed into reusable components. The essential interfaces like farmer dashboards, customer dashboards, administer panels, and product pages are displayed as separate React units. Common components like dashboard layouts, status indicators and notification systems are reusable and make it easier to reduce redundancy and enhance code maintainability. React hooks manage the state of an application, whereas React Router allows navigating between pages without reloading a whole page.

6.5 Node.js (Backend Runtime)

The backend runtime environment is Node.js, where all the operations on the server side are processed. It handles API requests like authentication of users, product management, order handling and submissions of reviews.

Its non-blocking nature enables the system to support several requests at a time, which is critical to a platform that has several users in action. Node.js also works well with JSON, so it will not be difficult to communicate between the frontend and the backend.

6.6 Express.js (Backend Framework)

The backend is structured with Express.js that handles API routing. It offers a clear and structured method to establish paths of various modules including authentication, operations of the farmer, interactions with customers and administrative facilities.

Authentication, role-based access control, and request validation are features that are implemented by middleware. This makes sure the backend is secure, modular and simple to maintain.

6.7 MySQL Database

All data stored in the platform is in MySQL. This covers user accounts, product listings, orders, reviews and recipes among other information. MySQL is relational and thus it is appropriate in applying interrelated data, e.g. between users and orders and products. Structured queries and constraints are some of the features that can be used to maintain data integrity and provide an efficient access to data.

6.8 JSON (Data Exchange Format)

As a standard form of communication between the frontend and backend, JSON is employed. The API requests and responses are in the form of a lightweight and easy to process data in the form of JSON. It makes debugging easier and compatibility between various parts of the system a certainty.

6.9 JWT (JSON Web Tokens) Authentication

Implementation of secure authentication in the platform is done using JWT. Upon the user logging in, a token is created and it is stored on the client-side. This is a token that is added to the further API calls that can identify the user. The token is validated by middleware functions, which perform role-based access control, so that users are only allowed to access functionality pertinent to their role. This method offers security and does not have server-side sessions.

6.10 Multer (File Upload Handling)

File uploads are processed using multer in the platform. It deals with any type of documents like identity proofs, profile pictures and product pictures uploaded by the user. The system provides a secure way of storing files, assigns a unique naming of files to avoid conflicts and imposes a limitation on the size and type of files to ensure the integrity of the system.

6.11 bcryptjs (Password Security)

The passwords of users are secured with the help of bcryptjs that hashes the passwords and then store them. The passwords typed during a login are compared to the stored hashes and the passwords themselves are not revealed. This improves the general security of the platform, particularly to those users who might not be very conscious of practices of digital security.

6.12 Visual Studio Code

As the main development environment, Visual Studio Code is adopted. It has functionalities like syntax highlighting, debugging, and built-in terminal support, which enhance the development speed. The frontend, backend and database operations can be handled in parallel and as a result, the development process is streamlined.

6.13 Browser Developer Tools

Test and debugging (Browser developer tools) are heavily used. They assist in tracking API requests, checking UI components, and real time error detection. These tools are essential in making sure that the frontend and the backend communicate with each other as well as to ensure that the system works in the manner that the system is intended to work.

6.11 Summary of Tools

Table 6.1 Summary of Tools Used

Tool	Purpose	Category
HTML	Page structure	Frontend
CSS (Inline)	Styling and UI consistency	Frontend
JavaScript	Interactivity and API handling	Frontend
React.js	Dynamic UI components	Frontend
Node.js	Backend runtime	Backend
Express.js	Routing and API management	Backend
MySQL	Data storage	Database
JSON	Data exchange format	Data
External API	Optional data integration	Data Source
VS Code	Development environment	IDE
GitHub	Version control	Repository

Chapter 7

Sustainability, Social, Legal and Ethical Aspect

This chapter presents the Millets Value Chain Platform in the technical point of view and its impact on the society, legal compliance, ethical responsibility, sustainability and safety. The modern software systems are not only anticipated to work well but also be responsible, inclusive, regulation compliant and long term sustainable. This chapter evaluates how the project is fulfilling these general expectations.

7.1 Social Aspects

The platform is developed with the main aim of serving the farmers, enhancing transparency in the supply chain, and encouraging people to consume millets. It is not a usual technical system but a socially efficient solution that assists in developing agriculture and being conscious.

7.1.1 Millets Awareness and Promotion of Agriculture

Millets are the nutritious and sustainable crops, but their awareness and consumption are low among the farmers and consumers (Dayakar Rao et al., 2017). In most instances, farmers do not have direct access to markets, and consumers are not well informed on the nutritional and environmental advantages of millets (Ritu and Mercykutty, 2025).

The site seeks to overcome these issues by raising awareness on various varieties of millets and their health advantages. It allows farmers to reach directly into markets, enhancing their chances of better income and lessening their reliance on intermediaries (Sangappa et al., 2020).

It also encourages sustainable farming that is environmentally friendly as it supports the production of climate-resistant crops, like millets (FAO, 2023). The platform offers farmers and consumers greater interaction by incorporating information, services, and a digital marketplace into a single platform, making the millet ecosystem stronger.

7.1.2 Inclusion and Accessibility

The platform will be made to be accessible by a large number of users and they include farmers, suppliers, consumers and administrators. The interface is user-friendly, intuitive and simple, making it easy to use even to those with little or no technical expertise. Its mobile-friendly design and minimal data usage provide rural people with special attention to accessibility over long distances, which is why it is applicable to the areas with low levels of internet connectivity.

Other additions that the system is engineered with regards to future upgrades include the multilingual support and voice-assistance features to enhance inclusivity. These characteristics will see to it that rural and semi-urban users can successfully utilize the platform and enjoy its benefits.

7.1.3 Educational Value

The platform can be used as an educational tool as it offers comprehensive information about various kinds of millets and their advantages (Dayakar Rao et al., 2017). It also describes supply chain processes which the users can use to know how agricultural products pass through production and to the consumer (Trienekens, 2011).

Moreover, the platform enhances the level of transparency because market pricing and associated information is displayed in a convenient format, enhancing the decision-making process (Ahmed et al., 2021).

With an integration of these features, the system allows students, researchers and agricultural stakeholders to learn how digital technologies could improve the traditional agricultural systems (Jayalath et al., 2021).

7.1.4 Farmer Empowerment and Social Equity

The platform helps provide equal opportunities to all the stakeholders, and especially farmers are empowered. It facilitates farmer-to-market links, thus minimizing reliance on middlemen and enhancing price discovery.

It also gives the same visibility to the small-scale farmers such that they are not overshadowed by the large-scale producers. The platform is inclusive of agricultural ecosystem by promoting diversity in agriculture and promoting the growth of millet in various regions. These actions will raise the level of farmer income and decrease inequality within the agricultural market.

7.2 Legal Aspects

Developing a platform to handle user data, transaction and external integrations should be conducted with due legal provisions.

7.2.1 Data Protection and Privacy

The site makes sure that the data of users is managed in a responsible and secure manner. It adheres to standards like gathering the minimum amount of information, keeping credentials in a safe place and not abusing or sharing personal information unnecessarily.

There are secure authentication mechanisms to safeguard the identity of the user and to guarantee data privacy.

7.2.2 Open-source Technologies

The site is developed on open-source platforms like Node.js, React.js, MySQL and CSS. These tools are employed in accordance with their own licences, so that the system does not contravene any legal or intellectual property law.

7.2.3 API Usage Compliance

The external APIs, like those of the market data or logistics, are done responsibly. The site complies with API usage guidelines by preventing an overload on API usage, depriving it of storing restricted information, and offering appropriate attribution where necessary. This guarantees ethical and legal assimilation of third-party services.

7.2.4 Content and Marketplace Regulation

The site makes all the information about products accurate and valid. It does not engage in misrepresentation and makes sure that the transactions in the marketplace obey the fundamentals of digital commerce. Such practices assist in earning trust of the users and upholding transparency within the system.

7.3 Ethical Aspects

The ethical issues are paramount when it comes to agricultural data, information about farmers, and the interactions on the marketplace.

7.3.1 Equitable representation of Farmers

The platform will also be fair to all the farmers without favor or discrimination. It does not favor large producers at the expense of small-scale farmers and makes sure that the information about products is reported without any manipulations.

7.3.2 Accuracy of Information

Any data concerning the type of millet to be used, prices, and supply chain operations will be checked and then shown. This will minimize chances of misinformation and the user will depend on the platform to get the correct data.

7.3.3 Avoidance of Bias

The system is inclusive as it offers equal chances to farmers in diverse regions and encourages other types of millets. It provides exactly the same visibility to every seller, thus ensuring fairness in the platform.

7.3.4 Ethical Use of Technology

The platform is meant to be responsible in using technology, not misuse the user data, and be transparent in its operations. It lacks design practices that are manipulative and value user trust and fairness.

7.4 Sustainability Aspects

The site is exceedingly interested in sustainability, both technology-wise and in the farming sector.

7.4.1 Digital Sustainability

The system is planned to be lightweight and efficient with optimized frontend technologies and scalable backend. This saves on resource use and enhances overall performance, which helps in energy efficiency.

7.4.2 Agricultural Sustainability

Millets are inherently sustainable crops since they need less water, thrive in unfavorable conditions, and enhance soil well-being (FAO, 2023). The platform encourages sustainable agriculture by encouraging cultivation of millet, which is environmentally friendly.

7.4.3 Long-Term Maintainability

The platform has a scalable modular design enabling easy extension and addition of new functionalities. This guarantees durability and flexibility to the technological changes in the future.

7.4.4. Minimization of Physical Resources

The platform will decrease reliance on paper-based systems and manual tracking by digitizing records and processes. This helps in environmental sustainability and enhances efficiency.

7.5 Safety Aspects

Safety encompasses digital security as well as user protection.

7.5.1 Secure Digital Environment

The platform does not allow irrelevant and harmful content to be displayed, which guarantees a secure browsing experience and ensures the management of user interactions.

7.5.2 Secure Backend Architecture

Security measures such as protected APIs, secure database connections, and input validation are implemented to safeguard the system against cyber threats and unauthorized access.

7.5.3 Prevention of Misinformation

The information on the platform is all checked to guarantee the accuracy of information on the product, price, and supply chain tracking. This assists in ensuring reliability and trust by the user.

7.5.4 Safe External Integrations

External services are only used with the help of reliable ones. Anything that is unsafe or unreliable is avoided to make sure that the system is intact.

Chapter 8

Conclusion

The main aim of the Millets Value Chain Platform has been realized. The project manages to integrate fragmented agricultural processes into one, digital and interactive system. It links farmers, suppliers and consumers in a centralized system enhancing transparency and efficiency in the millet supply chain. The flexibility, constant improvements, and risk management were made possible by the hybrid Agile approach taken in the development. It enabled the system to develop based on the feedback of iterations and at the same time ensuring that all functional and technical requirements that were found during the analysis phase were effectively addressed.

8.1 Summary of Approach and Results

It has been created systematically and the two most important aspects included a backend that was built in Node.js and MySQL database and a frontend that is built in React.js and HTML, CSS and JavaScript. The unveiling of a robust backend architecture that would accommodate dynamic data on farmers, products, transactions, and phases of the supply chain were among the key successes of the system. A smooth connection between the frontend and backend components was achieved through the use of REST APIs. Also, fallback mechanisms were provided to manage failures of external data sources to maintain the reliability of the system and its continuous operation. Testing phase was used to ensure that the system is stable and that it functions in normal conditions. It was able to implement and test all the key features like farmer registration, product listing, supply chain tracking, and marketplace interactions. The platform showed good functionality and integration testing which confirmed its usefulness as a digital supply chain solution. However, during the performance testing when the network was being tested on slower networks, delays were experienced due to massive loads of data and unoptimized media components. It implies that the system architecture is robust with the need to do more optimization to enhance the performance in low-bandwidth conditions.

8.2 Achieving Project Goals

The project was able to achieve all its set goals. The issue of fragmented agricultural information and supply networks that are not linked was addressed by establishing a central platform that brings together all the stakeholders. The system is interactive and user friendly whereby farmers are allowed to post products, users to browse and buy millets and administrators to manage the system effectively. The supply chain tracking, dashboards, and structured data management features were more usable and engaging. The open-source technology stack was employed to make sure that the project was developed at no cost, which was cost-effective and scalable. Moreover, the platform contributes to sustainable development by promoting the use of millet, supporting farmers, and improving farming techniques, which is in line with the broader goals of food security and economic development.

8.3 Future Recommendations

Based on the analysis and observations, the following recommendations are offered to enhance the platform and make it sustainable in the long-term:

Enhanced Accessibility Features:

Text-to-Speech (TTS) and multilingual support can be added to the platform to improve it. This will ensure the system becomes more user friendly to the farmers and other users that have different language and educational backgrounds and rural locations.

Performance Optimization:

The solution to the performance issues should include image and data optimization techniques. This involves media file compression, API response optimization and better frontend rendering to lessen load times to less than 3 seconds even over slow networks.

Administrative Management System:

A management panel should be created that will be able to control the content of the database safely and conveniently such as product listing, user information and transaction history. This will ensure that the platform is maintained in good condition and current in the long term.

Chapter 9

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Base Paper

From References the mainly referred paper:

[1] C. Ritu and M. J. Mercykutty, "Value chain dynamics of millet cultivation among tribal farmers in Attappadi, Kerala," International Journal of Agriculture Extension and Social Development, vol. 8, no. 12, pp. 187–190, 2025.

APPENDIX

i. Publications

- Submission letter for conference paper

2nd International Conference on Distributed Systems, Computer Networks and Cybersecurity (ICDSCNC-2026) : Submission (714) has been created. Inbox ×

Microsoft CMT <noreply@msr-cmt.org> 11:27 AM (15 minutes ago)
to me

Hello,

The following submission has been created.

Track Name: ICDSCNC2026

Paper ID: 714

Paper Title: Millets Value Chain Platform

Abstract:
Millets have gained renewed importance due to their high nutritional value and environmental sustainability. However, the millet value chain in India and other developing economies remains fragmented, with limited market access, lack of price transparency, poor coordination, and low digital integration. Smallholder and tribal farmers face low bargaining power, weak certification systems, and dependence on intermediaries, which restricts their income and market participation.

This paper proposes a Digital Millet Supply Chain Platform that connects farmers, processors, consumers, and administrators on a single system. It enables direct transactions, seller verification, raw material sourcing, sale of processed goods, traceability, price transparency, waste-to-value (ACM) utilization, and online order tracking. The platform is based on modular architecture, role-based dashboards, and digital monitoring to support a sustainable value chain.

The approach integrates value chain optimization, market insights, and consumer-focused strategies to create a technology-driven ecosystem. It aims to enhance farmer income, support startups, ensure quality, improve transparency, and build a sustainable and efficient millet economy.

Created on: Tue, 21 Apr 2026 05:57:36 GMT

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Secondary Subject Areas: Not Entered

Submission Files:
Research Paper CSE_24 - MilletValueSupplyChain.docx (69 Kb, Tue, 21 Apr 2026 05:57:29 GMT)

Figure A.1: Conference Paper Submission

ii. Research paper: Similarity Report

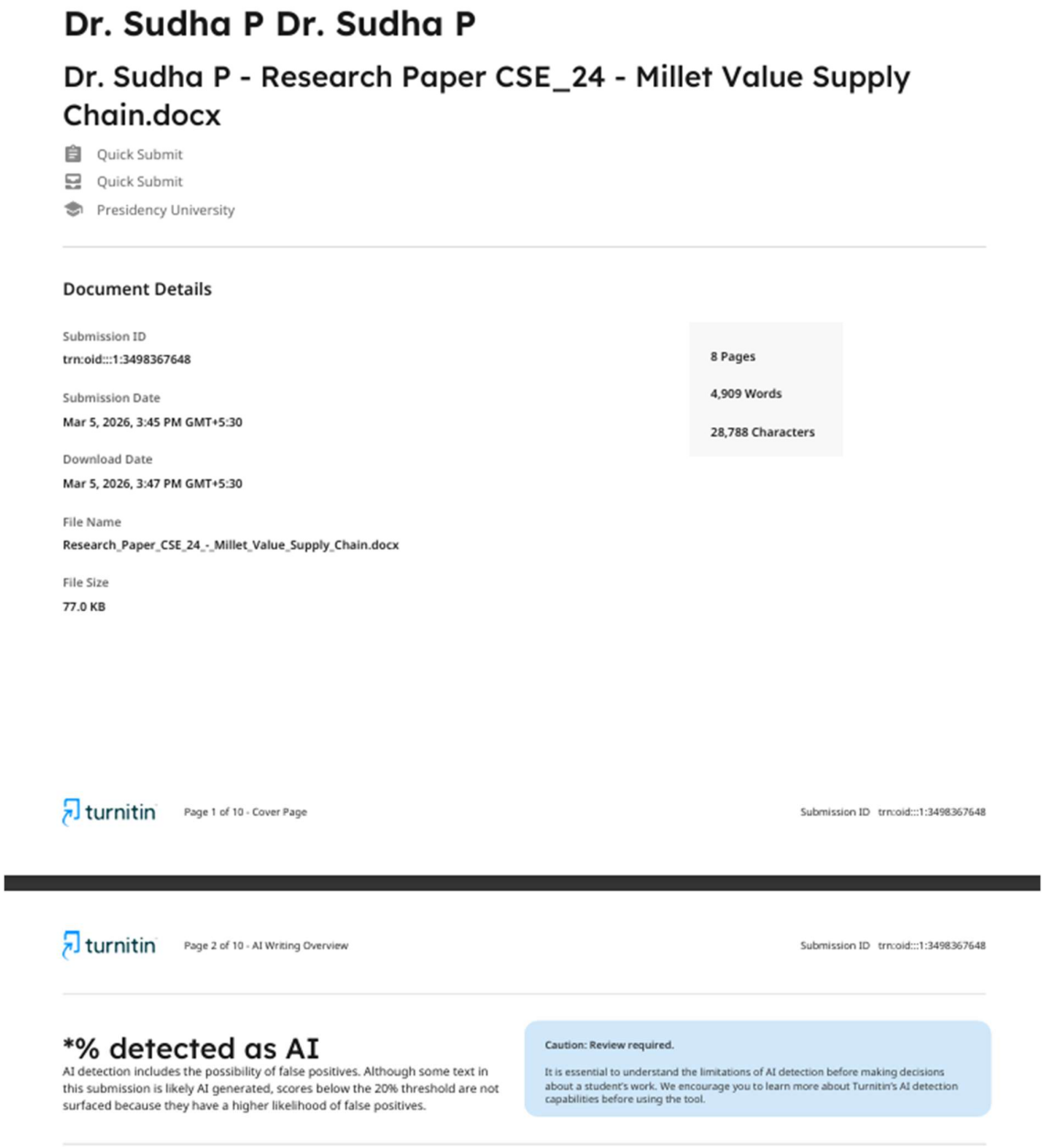

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Figure A.2: Research Paper - AI Similarity Report

iii. Live Project Demo

• GitHub: https://github.com/INDUJA2304/Millets-Value-Chain-Platform/commits?author=INDUJA2304

• Live Demo:

https://drive.google.com/drive/folders/1YF6FEDIKM7m_LCtGPBMw2TAK73QJXalb

iv. Few Images of Project

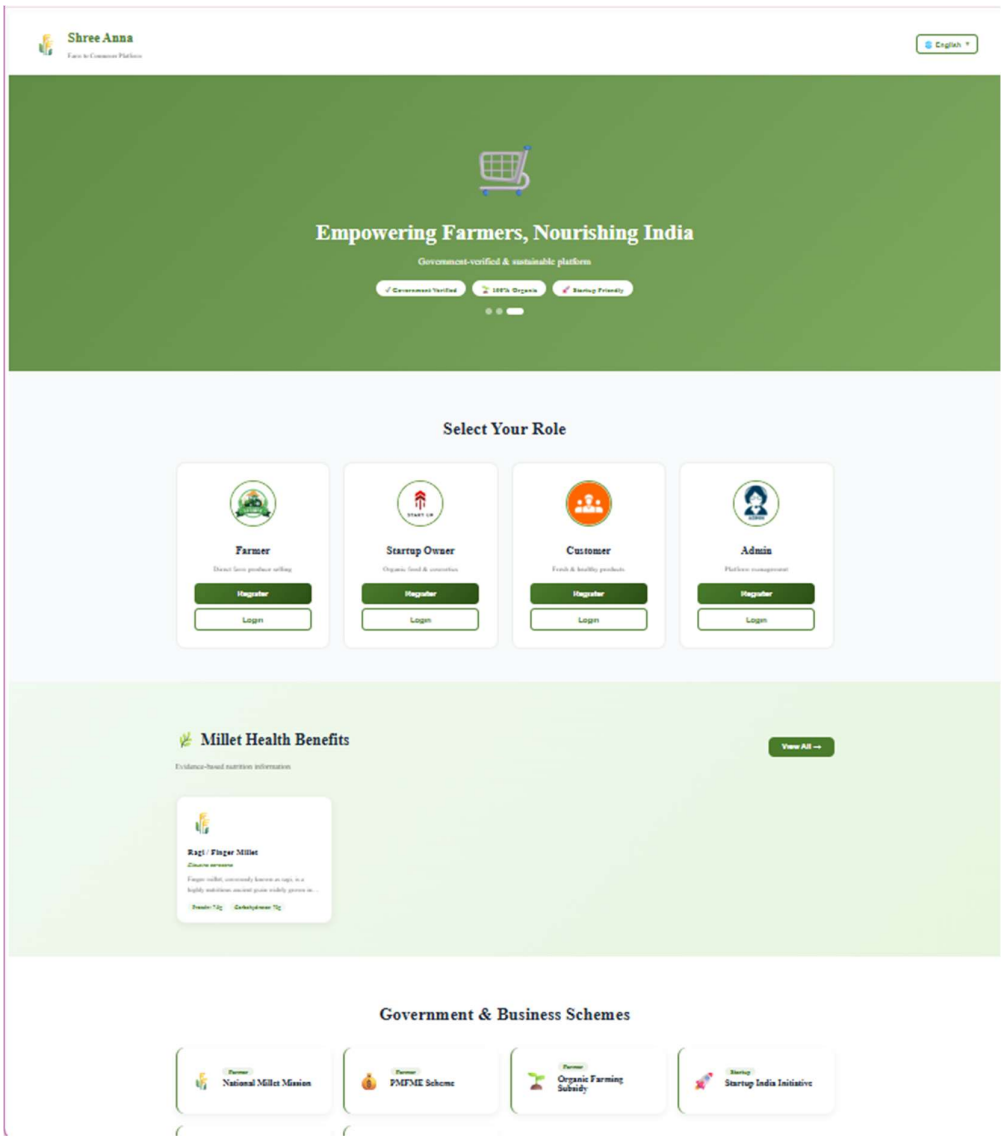

Figure A.3: Homepage

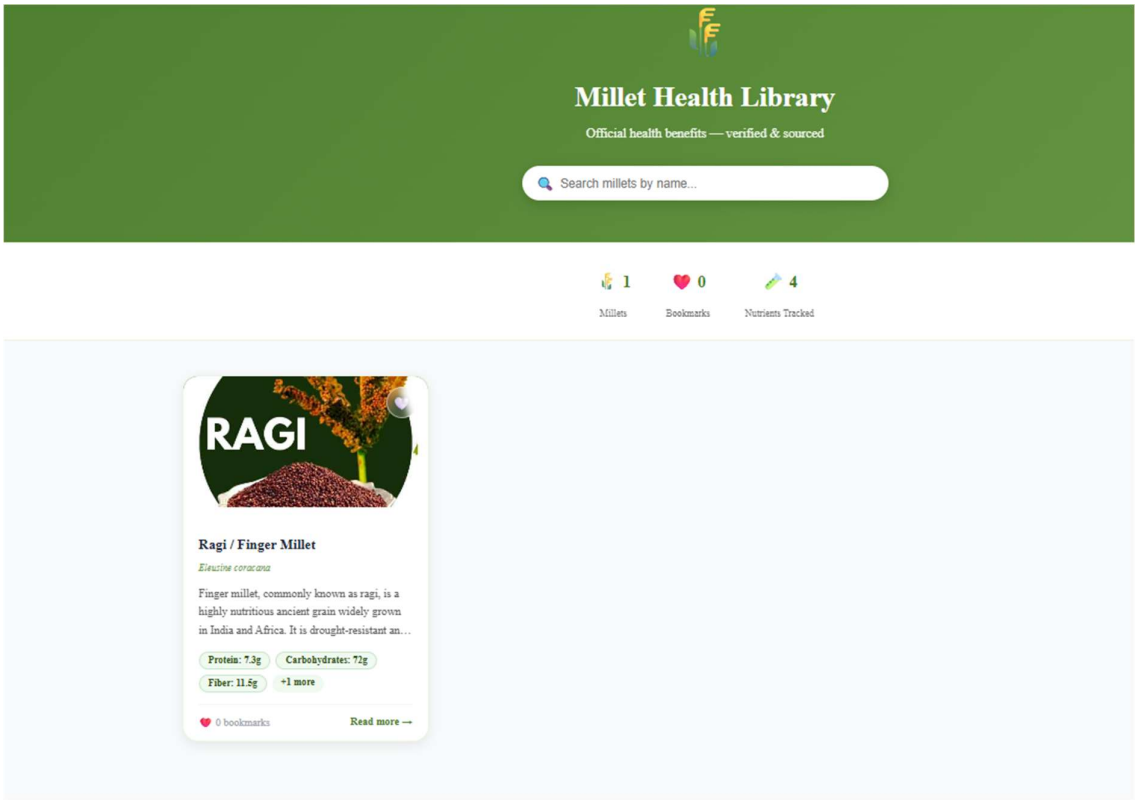

Figure A.4: Millet Library

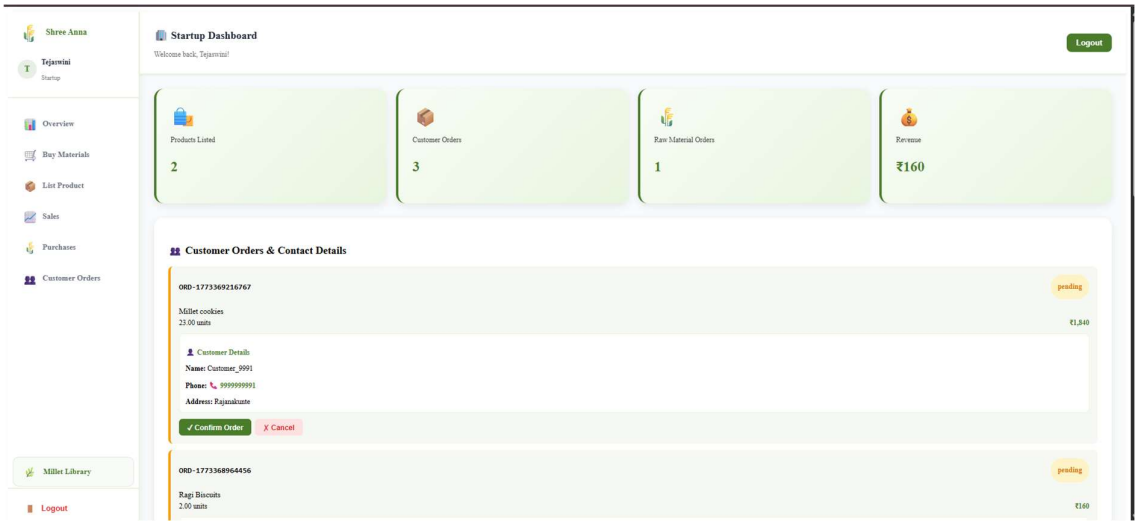

Figure A.5: Startup Dashboard

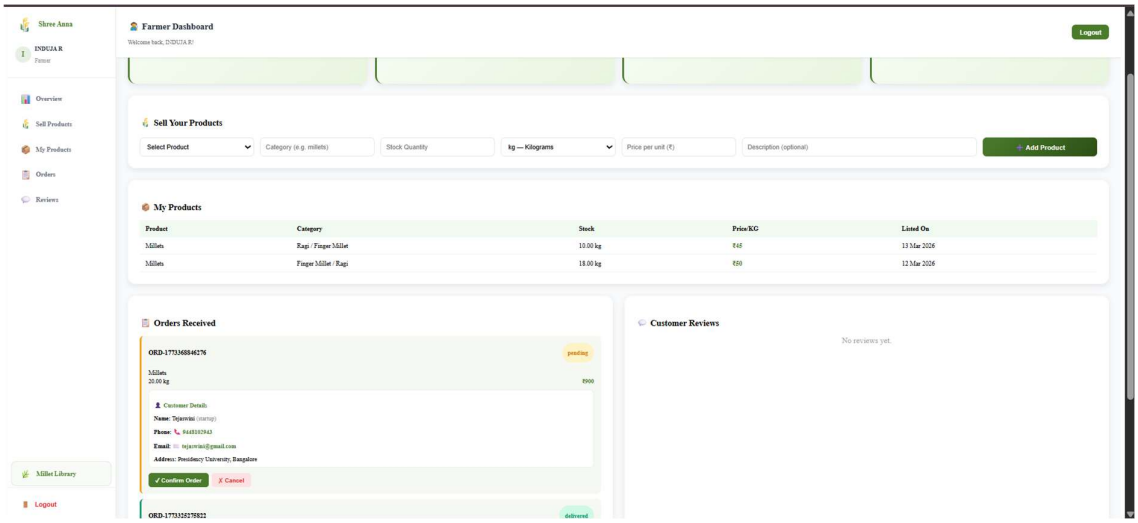

Fig A.6: Farmer Dashboard

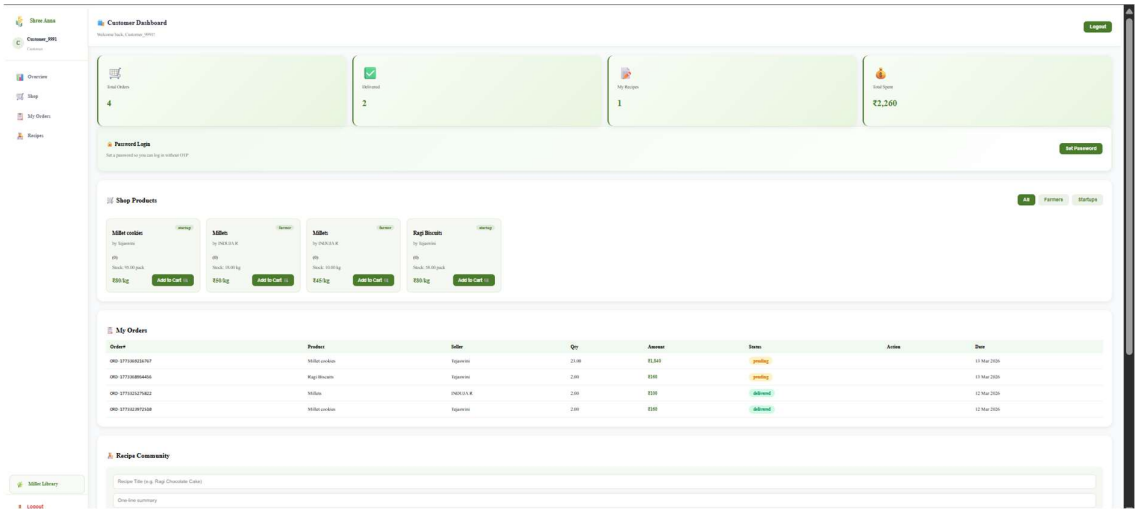

Figure A.7: Customer Dashboard

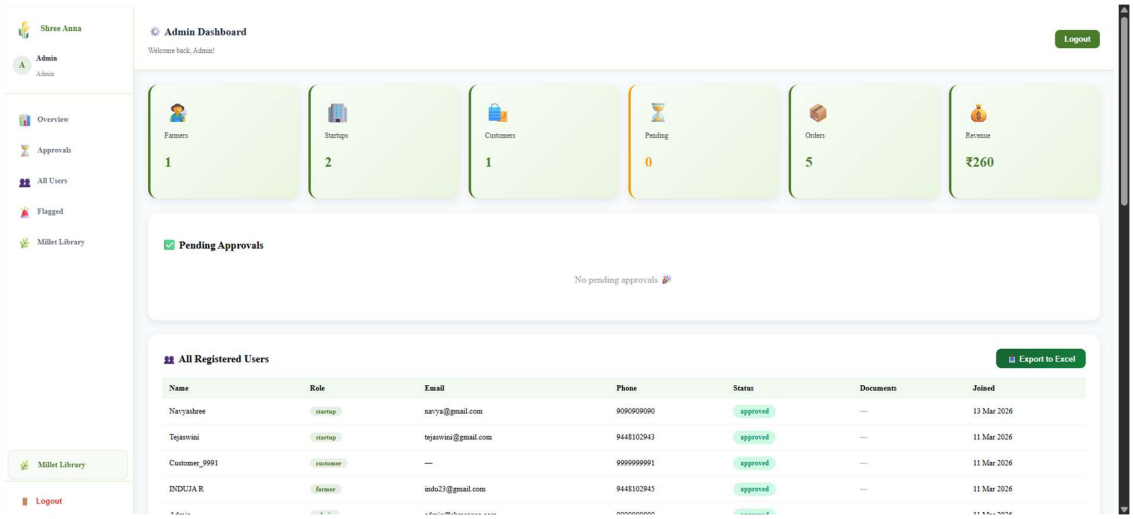

Figure A.8: Admin Dashboard

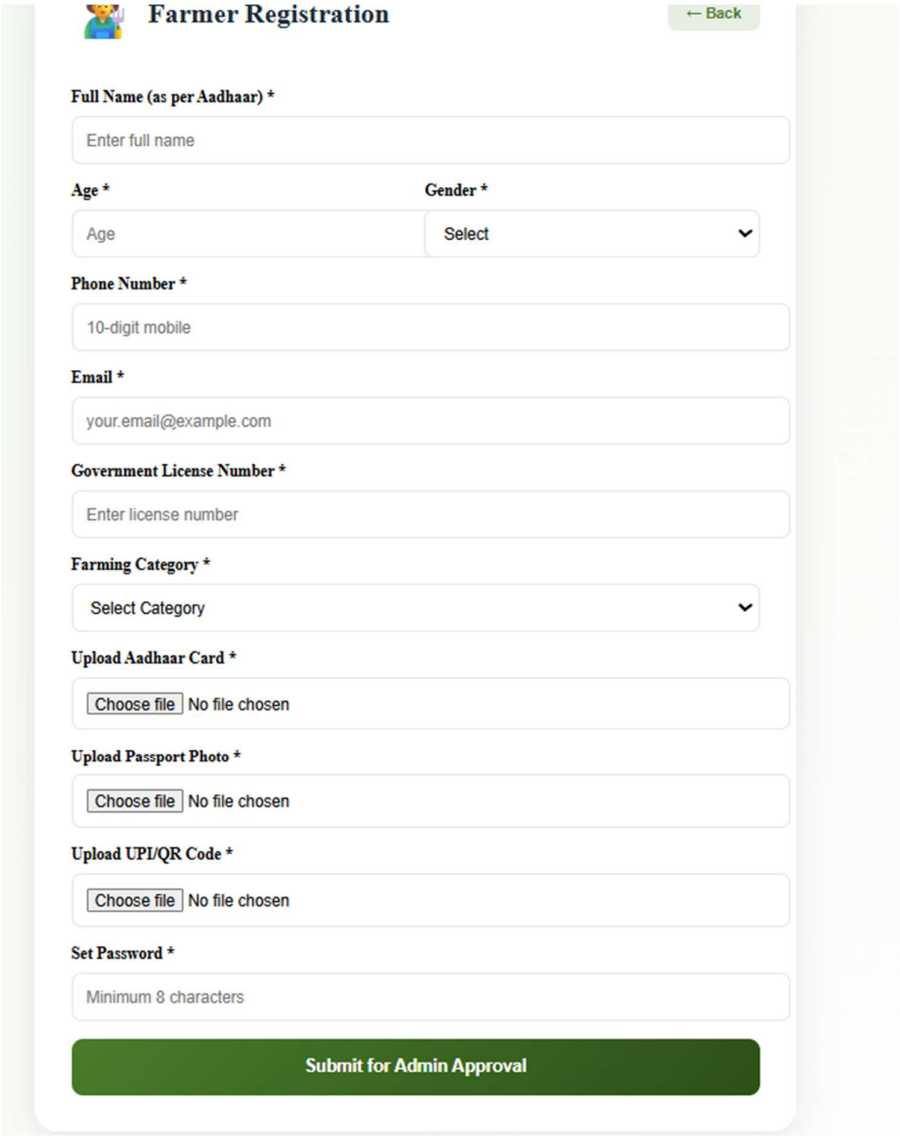

Figure A.9: Farmer Registration